BA/MIS 649 Section 1: Business Analytics
Final Project: Business Analytics on London Bike Sharing

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Part I: Introduction

Background of the Business Problem

Bike-sharing systems have become essential to urban transportation networks, providing environmentally friendly and convenient alternatives to traditional commuting. For cities like London, understanding the demand for bike rentals is crucial to ensuring efficient operations, proper resource allocation, and adequate infrastructure planning. The challenge is predicting and managing this demand, as various temporal and weather-related factors influence it.

Regression Problem Introduction

Business Question
What factors most significantly influence the hourly demand for bike rentals in London?

Dependent Variable (DV)
The dependent variable (DV) is “cnt”, the number of bikes rented per hour.

Potential Independent Variables (IV)
The initial potential independent variables (IVs) are “t1”(temperature), “hum”(humidity), wind_speed, weather_code, is_holiday, is_weekend, season, and hour. These are the IVs from the original dataset. (We create more variables at a later time.)

Importance for Business Practice
Accurate demand predictions help operators balance bike supply, reduce station shortages or surpluses, and optimize staffing and logistics.

Hypothesis
H1: Higher temperatures are associated with increased bike usage.
H2: Rental counts are lower during holidays and bad weather conditions.

Classification Problem Introduction

Business Question
Based on weather and time-related factors, can we predict whether the bike demand for a given hour will be high or low?

Dependent Variable (DV)
A binary variable, high_demand, will be created once we identify an appropriate threshold.

Potential Independent Variables (IV)

The potential IVs are the same as the above regression problem. They are “t1”(temperature), “hum”(humidity), wind_speed, wather_code, is_holiday, is_weekend, season, hour.

Importance for Business Practice
Predicting high-demand periods allows better allocation of bikes and enables businesses to be proactive in their logistical adjustments.

Hypothesis
H1: High demand is more likely during weekdays during commuting hours (7-9 AM, 4-6 PM).
H2: Low temperature and high humidity reduce the likelihood of high demand.

Part II: Data Preparation and EDA

Data Source

Source: Kaggle – London Bike Sharing Dataset
Number of Observations: 17,414
Each observation represents one hour of bike rental activity in London between 2015 and 2017.

Variable	Description	Type
timestamp	Date and time of the record	Qualitative
cnt	Count of total bike rentals per hour	Quantitative (continuous)
t1	Temperature in Celsius	Quantitative
t2	Feels-like temperature in Celsius	Quantitative
hum	Humidity (%)	Quantitative
wind_speed	Wind speed in km/h	Quantitative
weather_code	Weather condition code: 1 = Clear; mostly clear but has some values with haze/fog/patches of fog/ fog in vicinity 2 = scattered clouds / few clouds 3 = Broken clouds 4 = Cloudy 7 = Rain/ light Rain shower/ Light rain 10 = rain with thunderstorm 26 = snowfall 94 = Freezing Fog	Qualitative (categorical)
is_holiday	Indicator if the day is a holiday (0 = no, 1 = yes).	Qualitative (binary)
is_weekend	Indicator if the day is a weekend (0 = no, 1 = yes)	Qualitative (binary)
season	Category field meteorological seasons: 0-spring; 1-summer; 2-fall; 3-winter	Qualitative (categorical)

It is important to note that the original data set does not include a binary variable for "high_demand" relevant to our classification problem. After conducting an Exploratory Data Analysis (EDA), we will establish an appropriate threshold for defining what constitutes "high demand." We will then create this variable and add it to the dataset.

Exploratory Data Analysis

Check for Missing Values

Before we continue, we wanted to confirm if there were any missing values in our dataset.

```
# 1 Check for Missing Values
colSums(is.na(Bike))

   timestamp          cnt           t1           t2          hum
           0            0            0            0            0
  wind_speed weather_code   is_holiday   is_weekend       season
           0            0            0            0            0
```

There are no missing values in any column of the dataset. Each observation is complete and requires no deletion for modeling purposes.

Univariate Analysis

```
# 2 View Summary Statistics for All Variables
summary(Bike)

   timestamp                           cnt              t1
 Min.   :2015-01-04 00:00:00.00   Min.   :   0   Min.   :-1.50
 1st Qu.:2015-07-04 20:15:00.00   1st Qu.: 257   1st Qu.: 8.00
 Median :2016-01-03 15:30:00.00   Median : 844   Median :12.50
 Mean   :2016-01-03 22:31:00.56   Mean   :1143   Mean   :12.47
 3rd Qu.:2016-07-04 15:45:00.00   3rd Qu.:1672   3rd Qu.:16.00
 Max.   :2017-01-03 23:00:00.00   Max.   :7860   Max.   :34.00
       t2             hum           wind_speed     weather_code
 Min.   :-6.00   Min.   : 20.50   Min.   : 0.00   Min.   : 1.000
 1st Qu.: 6.00   1st Qu.: 63.00   1st Qu.:10.00   1st Qu.: 1.000
 Median :12.50   Median : 74.50   Median :15.00   Median : 2.000
 Mean   :11.52   Mean   : 72.32   Mean   :15.91   Mean   : 2.723
 3rd Qu.:16.00   3rd Qu.: 83.00   3rd Qu.:20.50   3rd Qu.: 3.000
 Max.   :34.00   Max.   :100.00   Max.   :56.50   Max.   :26.000
   is_holiday        is_weekend         season
 Min.   :0.00000   Min.   :0.0000   Min.   :0.000
 1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.000
 Median :0.00000   Median :0.0000   Median :1.000
 Mean   :0.02205   Mean   :0.2854   Mean   :1.492
 3rd Qu.:0.00000   3rd Qu.:1.0000   3rd Qu.:2.000
 Max.   :1.00000   Max.   :1.0000   Max.   :3.000
```

Summary of Key Variables (Output)

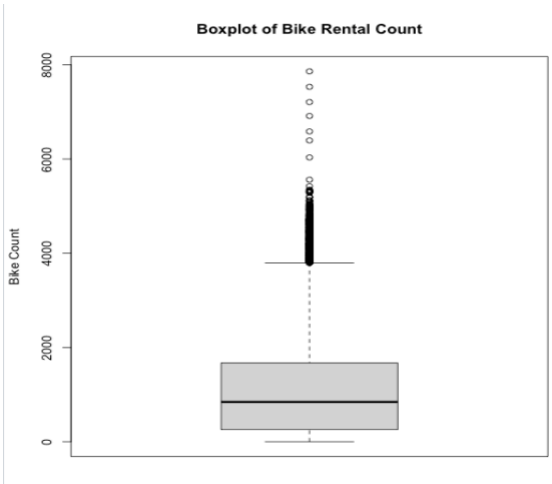
Cnt: Min is 0, 1st Quartile is 257, Median is 844, Mean is 1,143, 3rd Quartile is 1,672, and Max is 7,860. T1 has a range from -1.5°C to 34°C, and t2 has a range from -6°C to 34°C. Humidity has a range from 20.5% to 100%. Wind speed has a range from 0 to 56.5.

The cnt variable is right-skewed. The median is 844 while the mean is higher at 1,143, and the max value (7,860) suggests occasional very high rental spikes, likely during events or peak hours. Temperature (t1, t2) shows a relatively normal distribution centered around 12 °C. Humidity values span a wide range (20.5% to 100%) with a mean of about 72%, which indicates that weather conditions vary considerably. Categorical variables such as is_holiday, is_weekend, and season are encoded numerically and must be treated as factors once we do modeling.

Univariate Analysis for Bike Rental Count

Due to the Skewness in the Bike Rental Count, we created a boxplot to analyze the situation further.

```
# 3 Check for Extreme Outliers
boxplot(Bike$cnt, main = "Boxplot of Bike Rental Count", ylab = "Bike Count")
```



The boxplot for the "cnt" variable, which represents bike rental counts, shows a right-skewed distribution with a significant number of high-end outliers. The median rental count is just below 1,000, specifically at 844. The upper whisker of the boxplot extends nearly to 4,000, beyond which there are numerous outliers; the highest among them approaches 8,000 rentals in an hour, specifically reaching 7,860. These extreme values likely correspond to peak commuting hours or special events. They should be considered valid yet influential observations. Although they may impact modeling outcomes, they are not errors in data entry and have not been removed.

Univariate Categorical Analysis - Frequency Tables

Frequency tables were utilized for the categorical variables to show how data is distributed across categories.

```
# 4 Univariate Categorical Analysis - Frequency Tables
> table(Bike$season)

Spring Summer   Fall Winter
  4394   4387   4303   4330
> table(Bike$is_holiday)
```

```
    0     1
17030   384
> table(Bike$weather_code)

   1    2    3    4    7   10   26
6150 4034 3551 1464 2141   14   60
> table(Bike$is_weekend)

    0     1
12444  4970
```

The dataset is fairly evenly distributed across all four seasons, which allows for a reliable analysis of seasonal bike usage patterns, ensuring that no season is underrepresented. Holidays comprise only 384 out of 17,414 records, reflecting their typical occurrence in a given year. While the holiday data is limited, it may offer predictive value, as bike rental behavior could differ significantly on holidays compared to regular weekdays or weekends.

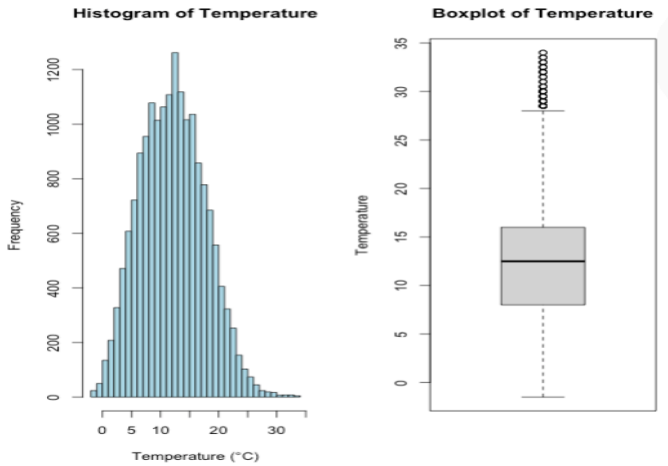
The table shows that most records are associated with clear to moderately cloudy weather conditions. Specifically, there are 6,150 observations for clear weather (1), 4,034 for scattered clouds (2), and 3,551 for broken clouds (3). Severe weather events, such as thunderstorms (10) and snowfall (26), are infrequent but do happen. These events may correspond with a decrease in bike demand.

The table also shows that about 28.5% of the observations occurred on weekends. This aligns with expectations, as weekends are two out of seven weekly days. The weekend data is sufficiently represented for reliable analysis of differences in bike rental behavior between weekdays and weekends.

Univariate Analysis for Temperature

Temperature could potentially be a strong predictor of demand; typically, warmer days likely mean more bike usage. To analyze temperature further, we created a histogram and a boxplot.

```
# 5 Univariate Analysis for Temperature
par(mfrow = c(1, 2))
hist(Bike$t1, breaks = 30, main = "Histogram of Temperature", col = "lightblue", xlab
= "Temperature (°C)")
boxplot(Bike$t1, main = "Boxplot of Temperature", ylab = "Temperature")
```

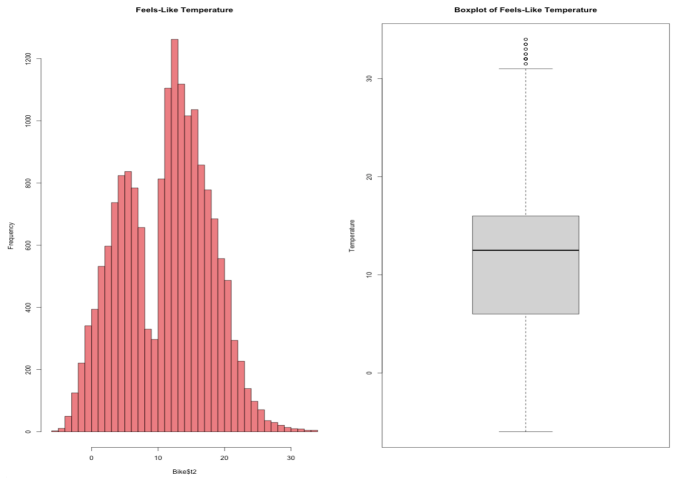


The temperature (t1) histogram displays a bell-shaped curve, indicating an approximately normal distribution centered around 12-13°C. Most observations fall between 5°C and 20°C, suggesting that moderate temperatures are the most common throughout the dataset. The smooth tapering on both ends reflects the natural seasonal variation in London.

The boxplot further supports the information shown in the histogram, with the median temperature being around 13°C. The interquartile range (IQR) is approximately from 8°C to 17°C. There are several outliers above 30°C, likely due to summer heatwaves. These high temperatures are expected and do not appear to be errors.

Next, we wanted to compare the "feels-like" temperatures (t2) with actual temperatures to identify discrepancies.

```
# 6 Feel-like Temperature
hist(Bike$t2, breaks = 30, main = "Feels-Like Temperature", col = "lightcoral")
boxplot(Bike$t2, main = "Boxplot of Feels-Like Temperature", ylab = "Temperature")
```



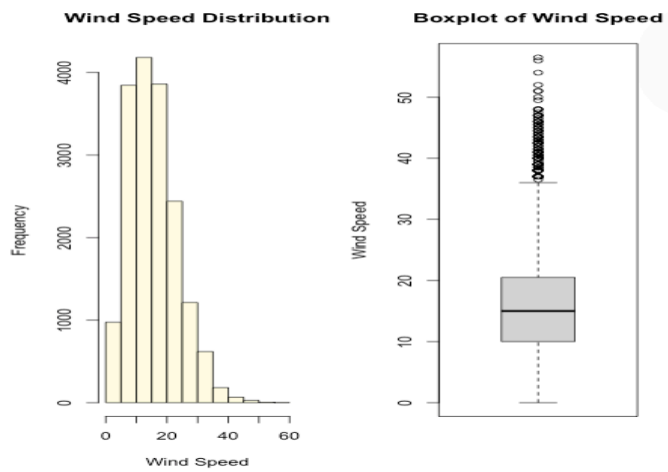
The distribution displayed here has two distinct peaks: around 8°C and between 13°C and 15°C. This indicates that London experiences two main temperature ranges—colder months and milder months. The dip between these peaks likely represents the transitional months. Overall, the distribution is mostly symmetrical around the main peak, but it has a longer tail on

the right side that extends to around 30°C, which reflects occasional heatwaves. The boxplot reinforces the trends observed in the histogram.

The bimodal nature of the temperature data suggests a seasonal influence, which could have a nonlinear effect on bike rental behavior. For instance, bike rentals may increase with rising temperatures but could decline again if it becomes too hot. Consequently, the analysis may need to consider potential transformation or interaction terms.

Univariate Analysis for Wind Speed

```
# 7 Univariate Analysis for Wind Speed
par(mfrow = c(1, 2))
hist(Bike$wind_speed, main = "Wind Speed Distribution", col = "lightyellow", xlab =
"Wind Speed")
boxplot(Bike$wind_speed, main = "Boxplot of Wind Speed", ylab = "Wind Speed")
```

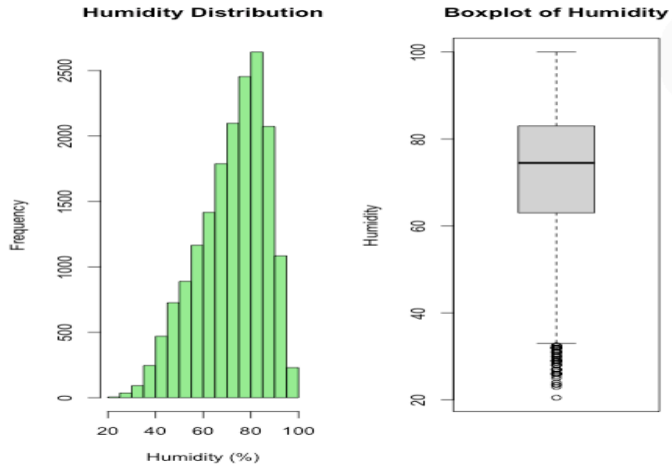


The distribution of wind speeds is right-skewed, meaning that most values are concentrated between 5 and 25 km/h. The most common wind speeds range from 15 to 18 km/h, which is classified as moderate. Although very high wind speeds are uncommon, they can exceed 50 km/h.

A box plot of the data illustrates the skewness and reveals the presence of outliers. The upper whisker extends to approximately 35 km/h, with several values plotted beyond that point. These high wind speeds may suggest stormy or extreme weather conditions, which could discourage bicycle users.

Univariate Analysis for Humidity

```
# 8 Univariate Analysis for Humidity
hist(Bike$hum, main = "Humidity Distribution", col = "lightgreen", xlab = "Humidity
(%)")
boxplot(Bike$hum, main = "Boxplot of Humidity", ylab = "Humidity")
```



The histogram shows a left-skewed distribution, with most observations grouped between 60% and 100%. A noticeable peak around 85-90% indicates that very low humidity levels are uncommon in this dataset.

The box plot reinforces this observation of skewness, displaying a concentrated upper range and a median humidity level between 75% and 80%. The lower whisker extends to approximately 40%, and several outliers are below 30%. These outliers likely represent cold, dry winter hours or specific weather events.

Addition of Variables

After analyzing the current dataset, we made several adjustments to improve its usability by incorporating time-based features. We extracted the hours of the day to identify peak hours, commute times, and daily patterns. Additionally, we included the day of the week to examine bike usage patterns by day further. We also added the month and year to account for seasonal trends and annual variations. Furthermore, we flagged specific commute hours, such as morning (7-9 AM) and evening (4-6 PM).

To address our classification problem, we created a binary variable called high_demand. Instead of relying on the mean or median, which would include too many average hours, we defined high demand as instances where rentals exceeded the 75th percentile (1,672 rentals). This definition ensures that "high demand" accurately represents the top 25% of operational load, aligning more closely with business needs such as bike redistribution and additional staffing.

```
# 9 Additional Variables
Bike$hour <- as.integer(format(Bike$timestamp, "%H"))
Bike$weekday <- as.factor(weekdays(Bike$timestamp))
Bike$month <- as.integer(format(Bike$timestamp, "%m"))
Bike$year <- as.integer(format(Bike$timestamp, "%Y"))
Bike$commute <- ifelse(Bike$hour %in% c(7, 8, 9, 16, 17, 18), 1, 0)

# 10 Use the Top 25% of Rental Counts for High Demand
quantile(Bike$cnt, 0.75)
Bike$high_demand <- ifelse(Bike$cnt >= 1672, 1, 0)
```

Once we added the above variables, our new list of variables are as follows:

Variable	Description	Type
timestamp	Date and time of the record	Qualitative
cnt	Count of total bike rentals per hour	Quantitative (continuous)
t1	Temperature in Celsius	Quantitative
t2	Feels-like temperature in Celsius	Quantitative
hum	Humidity (%)	Quantitative
wind_speed	Wind speed in km/h	Quantitative
weather_code	Weather condition code: 1 = Clear ; mostly clear but have some values with haze/fog/patches of fog/ fog in vicinity 2 = scattered clouds / few clouds 3 = Broken clouds 4 = Cloudy 7 = Rain/ light Rain shower/ Light rain 10 = rain with thunderstorm 26 = snowfall 94 = Freezing Fog	Qualitative (categorical)
is_holiday	Indicator if the day is a holiday (0 = no, 1 = yes).	Qualitative (binary)
is_weekend	Indicator if the day is a weekend (0 = no, 1 = yes)	Qualitative (binary)
season	Category field meteorological seasons: 0-spring; 1-summer; 2-fall; 3-winter	Qualitative (categorical)
hour	Hour of the day (0 to 23), extracted from timestamp	Quantitative (discrete)
weekday	Day of the week (Monday to Sunday), extracted from timestamp	Qualitative (categorical)
month	Month of the year (1= January,..., 12= December), extracted from timestamp	Qualitative
year	Calendar year (2015,2016,2017), extracted from timestamp	Quantitative (discrete)
commute	Indicator for commute hours (1=commute hour, 0=non-commute hour) (Commute hours 7,8,9,16,17,18)	Qualitative (binary)
high_demand	An indicator of high demand or not. The threshold is above 75%, which is 1672 (1 = high demand, 0 = not high demand)	Qualitative (binary)

We also ensured that we set certain variables as factors.

```
# 11 Set Variables as Factor
Bike$weather_code <- as.factor(Bike$weather_code)
```

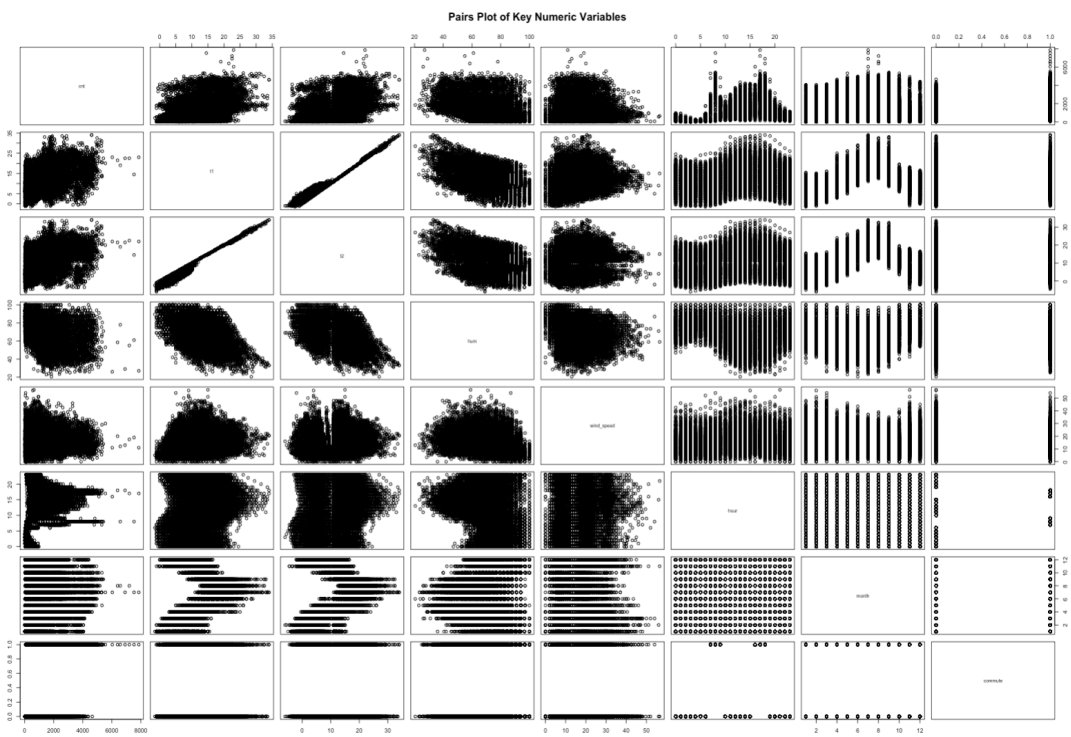
```
Bike$is_holiday <- as.factor(Bike$is_holiday)
Bike$is_weekend <- as.factor(Bike$is_weekend)
Bike$season <- factor(Bike$season, labels = c("Spring", "Summer", "Fall", "Winter"))
Bike$commute <- as.factor(Bike$commute)
Bike$high_demand <- as.factor(Bike$high_demand)
Bike$month <- as.factor(Bike$month)
```

Univariate Analysis for Humidity

Bivariate Analysis

To begin our bivariate analysis, we created a scatterplot matrix to visualize relationships and potential correlations between multiple numeric variables simultaneously. Since a pairs plot only works on numeric variables, we created a subset with only the numeric variables we wanted to visualize. We then generated the pairs plot as well as a Numeric Correlation matrix.

```
# 1 Generate Pairs Plot
pairs(Bike_numeric, main = "Pairs Plot of Key Numeric Variables")
# Automatically select only numeric columns
Bike_numeric <- Bike[, sapply(Bike, is.numeric)]
cor(Bike_numeric)
```




```
                     cnt           t1           t2          hum  wind_speed
cnt          1.00000000  0.38879845  0.36903479 -0.46290096  0.11629523
t1           0.38879845  1.00000000  0.98834422 -0.44778100  0.14547097
t2           0.36903479  0.98834422  1.00000000 -0.40349514  0.08840854
hum         -0.46290096 -0.44778100 -0.40349514  1.00000000 -0.28778917
wind_speed   0.11629523  0.14547097  0.08840854 -0.28778917  1.00000000
hour         0.32442347  0.16870842  0.15395623 -0.29565283  0.14179165
year         0.01004604 -0.03795949 -0.04497221  0.07244283 -0.09473938
                    hour          year
cnt          0.3244234663  0.0100460413
t1           0.1687084216 -0.0379594909
```

```
t2          0.1539562263 -0.0449722111
hum        -0.2956528314  0.0724428291
wind_speed  0.1417916512 -0.0947393832
hour        1.0000000000  0.0005424573
year        0.0005424573  1.0000000000
```

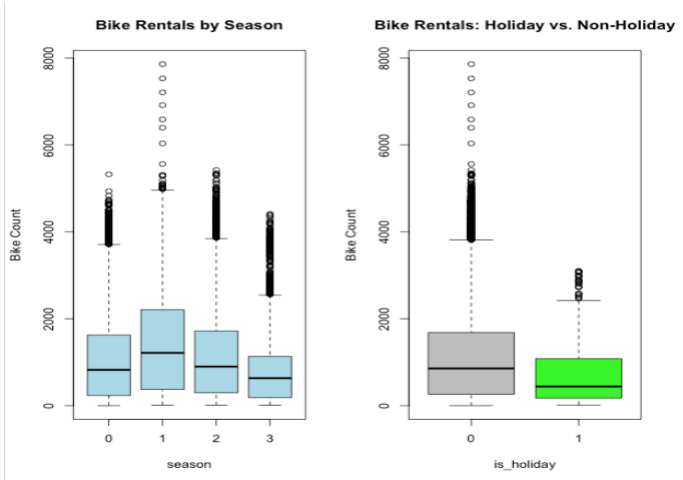
A very slight positive linear correlation was observed between temperature (t1) and feel-like temperature (t2) with bike rental counts (cnts), as expected. However, the trend appears slightly curved, suggesting that incorporating a quadratic term or a nonlinear transformation could enhance model accuracy. Humidity displayed a negative correlation with bike rentals. The feel-like temperature (t2) and actual temperature (t1) are nearly perfectly correlated, which indicates that we should only include one of these variables in our future model to avoid multicollinearity. The variable representing the hour of the day displays a nonlinear pattern in relation to bike rentals, which aligns with established commuting patterns. Additionally, there was no clear linear relationship between bike rental counts and either wind speed or month. This suggests that these variables may be more appropriate as categorical variables or in interaction terms.

We also examined a correlation matrix of all numeric variables to support our scatterplot analysis further. This analysis revealed a moderate positive correlation between bike rental counts (cnt) and temperature (t1), with a correlation coefficient (r) of approximately 0.39. This confirms that warmer temperatures are generally associated with higher bike rental counts. Similarly, the feels-like temperature (t2) shows a comparable correlation with cnt and is almost perfectly correlated with t1 ($r \approx 0.99$). This close relationship indicates that including both variables in the same model could result in multicollinearity. Therefore, we will include only one of these variables in our final models.

Additionally, the variable for humidity (hum) has a moderate negative correlation with cnt ($r \approx -0.46$), supporting the hypothesis that higher humidity may suppress demand for bike rentals. Other variables, such as wind speed, month, and year, show weak or near-zero correlations with cnt, reinforcing our earlier observation. These variables may need to be transformed, treated as categorical variables, or used in interaction terms rather than as direct linear predictors.

Boxplots for Rental by Season and Holiday

```
boxplot(cnt ~ season, data = Bike, main = "Bike Rentals by Season", col = "lightblue",
ylab = "Bike Count")
boxplot(cnt ~ is_holiday, data = Bike, main = "Bike Rentals: Holiday vs. Non-Holiday",
col = c("gray", "green"), ylab = "Bike Count")
```

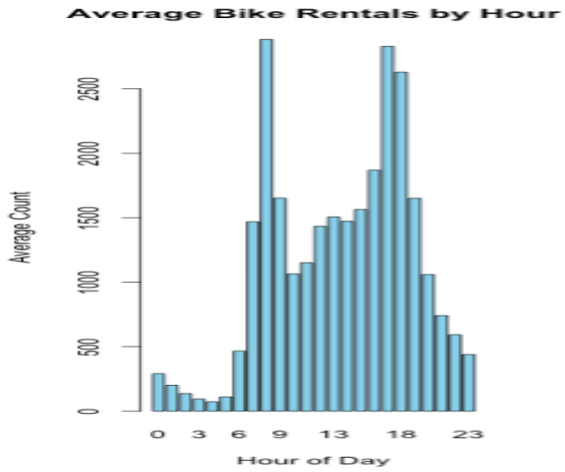


The data shows that Summer (1) has the highest median rental count and the widest interquartile range (IQR), indicating higher and more variable rent demand. Spring (0) and Fall (2) exhibit similar rental distributions. In contrast, Winter (3) has the lowest median rental count and a tighter IQR, reflecting consistently lower bike usage. All seasons have outliers, particularly Summer, which likely represent peak hours or special events that lead to unusually high demand.

The boxplot comparing bike rentals on holidays versus non-holidays reveals that bike usage is significantly lower on holidays. The median rental count decreases, and the overall spread of values becomes narrower, suggesting that holiday behavior is less varied and the demand for rentals is generally lower.

Bivariate Analysis of Hour and Average Count

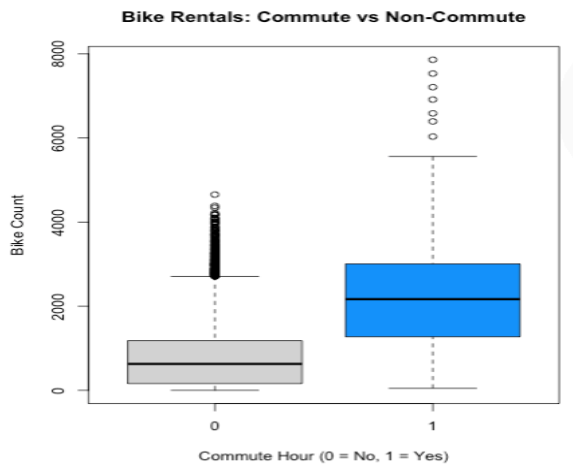
```
avg_cnt_by_hour <- tapply(Bike$cnt, Bike$hour, mean)
barplot(avg_cnt_by_hour, main = "Average Bike Rentals by Hour", col = "skyblue", xlab
= "Hour of Day", ylab = "Average Count")
```



The barplot of average bike rentals by hour shows clear spikes during traditional morning commute hours (7-9 AM) and evening (4-6 PM). These patterns indicate that bike sharing is heavily used for commuting, particularly during weekday mornings and evenings.

Bivariate Analysis of Commute and Count

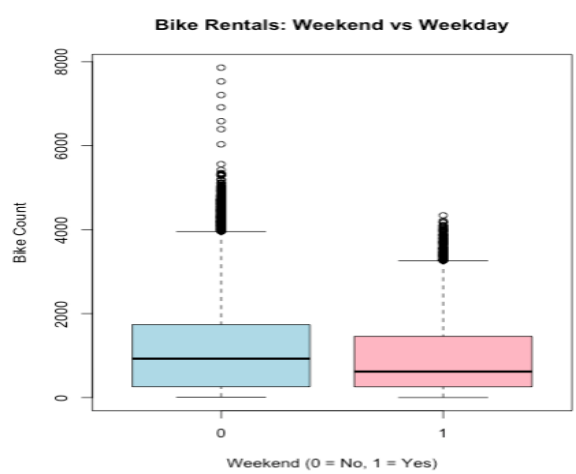
```
boxplot(cnt ~ commute, data = Bike, main = "Bike Rentals: Commute vs
Non-Commute",
        col = c("lightgray", "dodgerblue"), ylab = "Bike Count", xlab =
"Commute Hour (0 = No, 1 = Yes)")
```



The boxplot comparing bike rentals during commute and non-commute hours shows a clear difference. The median number of rentals is higher during commute hours, and the distribution is wider. This suggests that commute periods represent a significant portion of overall rental activity, which supports the inclusion of the commute variable in both regression and classification models.

Bivariate Analysis of Weekend and Count

```
boxplot(cnt ~ is_weekend, data = Bike, main = "Bike Rentals: Weekend vs Weekday",
        col = c("lightblue", "lightpink"), ylab = "Bike Count", xlab = "Weekend (0 =
No, 1 = Yes)")
```

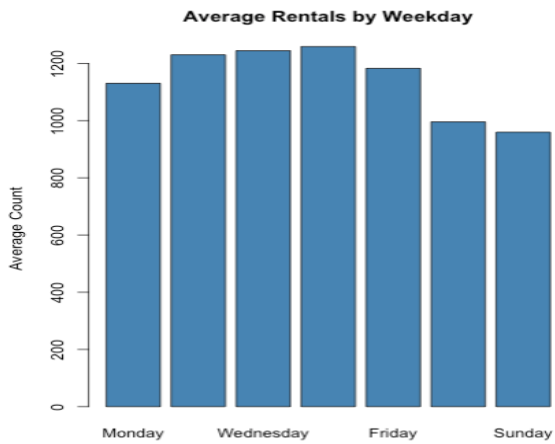


The boxplot shows that bike rentals are generally higher on weekdays than on weekends. This trend is likely due to weekday commuting habits, which aligns with other analyses of hourly

demand. In contrast, weekend rentals tend to be lower and more consistent. While the impact of weekend usage is not as significant as that of weekday commuting, the variable indicating whether it is a weekend can still help explain variations in demand. Therefore, it is important to include this variable in the modeling to effectively differentiate between weekday and weekend usage patterns.

Bivariate Analysis of Weekday and Average Count

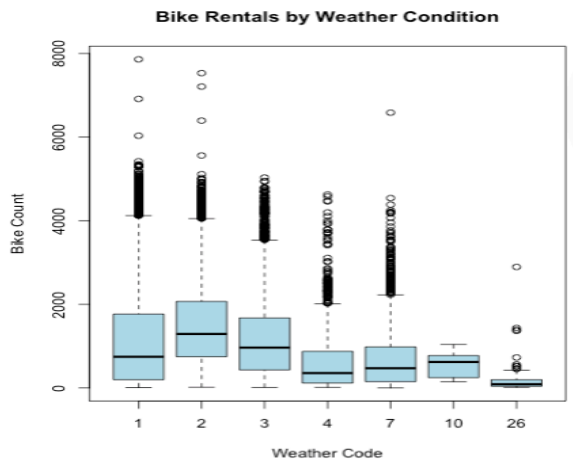
```
Bike$weekday <- factor(Bike$weekday,
                       levels = c("Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday", "Sunday"))
library(dplyr)
avg_by_weekday <- Bike %>%
  group_by(weekday) %>%
  summarise(avg_cnt = mean(cnt))

barplot(avg_by_weekday$avg_cnt, names.arg = avg_by_weekday$weekday,
        main = "Average Rentals by Weekday", col = "steelblue", ylab = "Average
Count")
```



The bar plot illustrating the average number of bike rentals by weekday shows a consistent increase in demand from Monday to Thursday, with Thursday featuring the highest average usage. There is a slight decline on Friday, followed by a more significant drop over the weekend. This pattern indicates that commuting and regular work schedules influence bike rental demand, supporting the use of weekday or related indicators in modeling efforts.

Bivariate Analysis of Weather Conditions vs. Bike Count

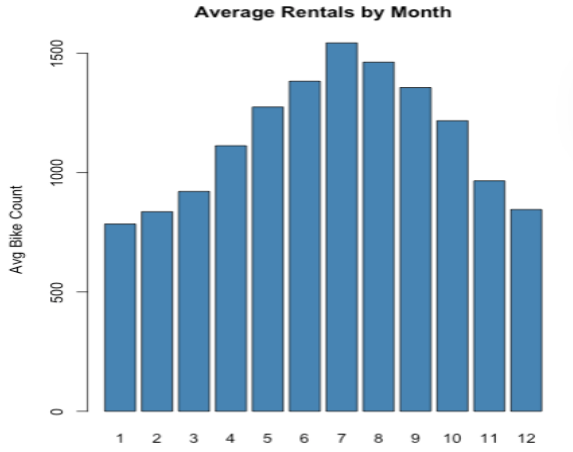
```
boxplot(cnt ~ weather_code, data = Bike, main = "Bike Rentals by Weather Condition",
        col = "lightblue", ylab = "Bike Count", xlab = "Weather Code")
```



The boxplot shows how hourly bike rental counts vary across different weather conditions. Codes 1-4 (Clear, scattered clouds, broken clouds, and overcast/cloudy) show higher and more variable usage. As weather severity increases, light rain (7), rain with thunderstorms (10), and snowfall (26), rental counts drop. Snowfall shows the lowest median and narrowest distribution, indicating a sharp suppression in bike demand during adverse weather conditions.

Bivariate Analysis of Average Rentals by Month

```
library(dplyr)
avg_by_month <- Bike %>%
  group_by(month) %>%
  summarise(avg_cnt = mean(cnt))
barplot(avg_by_month$avg_cnt, names.arg = avg_by_month$month,
        main = "Average Rentals by Month", col = "steelblue", ylab = "Avg Bike
Count")
```

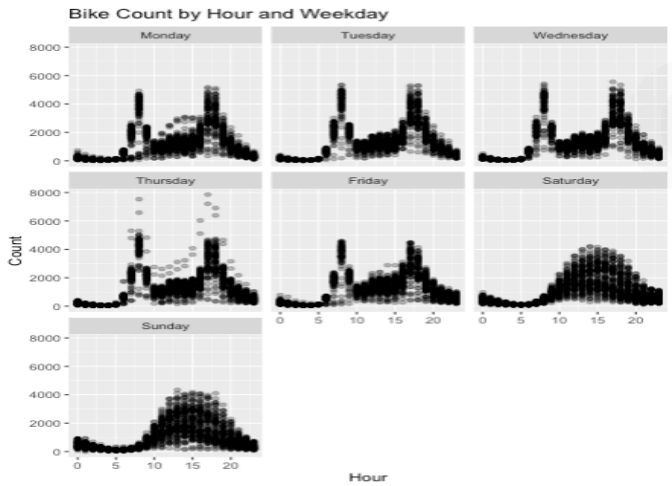


The barplot displays the average bike rentals per hour across each month. We can see that there is a strong seasonal trend. Rentals start relatively low in January- February. Demand steadily increases from March to July, peaking in July. After that, it gradually declines through December. This trend aligns with warmer weather and longer daylight hours, suggesting seasonal effects are a major driver of bike rental behavior. This shows that the month variable

should be included in modeling to account for cyclical trends affecting bike availability, planning, and operations.

Bivariate Analysis of Hour vs Count by Weekday

```
library(ggplot2)
ggplot(Bike, aes(x = hour, y = cnt)) +
  geom_point(alpha = 0.3) +
  facet_wrap(~ weekday) +
  labs(title = "Bike Count by Hour and Weekday", x = "Hour", y = "Count")
```

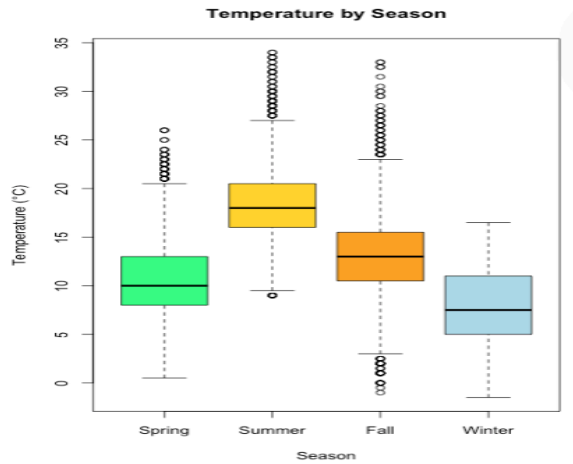


The scatter plot illustrates how bike rentals vary by hour of the day across different weekdays. As observed in previous analyses, there are distinct demand peaks on weekdays during the morning and evening commuting hours. In contrast, the pattern on weekends (Saturday and Sunday) is different; demand gradually increases throughout the morning and peaks mid-afternoon, showing no sharp spikes and reflecting a more leisure-based usage.

This analysis indicates that the variable "hour" interacts with "weekday," meaning that bike demand behaves differently depending on the day of the week.

Bivariate Analysis of Temperature vs Season

```
boxplot(t1 ~ season, data = Bike,
        main = "Temperature by Season",
        xlab = "Season", ylab = "Temperature (°C)",
        col = c("springgreen", "gold", "orange", "lightblue"))
```

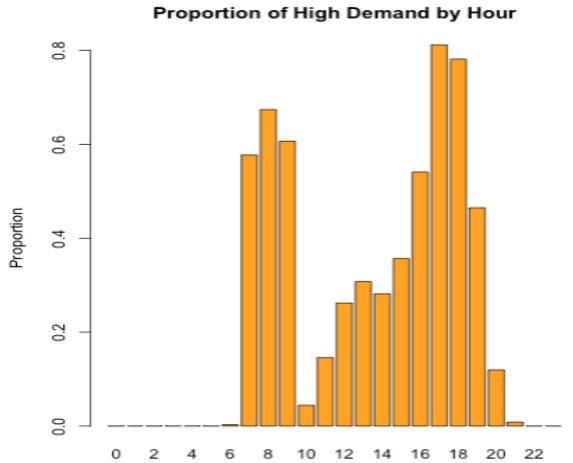


The temperature (t1) boxplot across seasons demonstrates a strong relationship between the two variables. As expected, temperatures are highest in the summer and lowest in the winter, with spring and fall falling in between. This overlap suggests that season and t1 are highly correlated. Since temperature is a continuous variable that captures daily variability more precisely, we included t1 in our final model and exclude season to avoid redundancy and improve interpretability.

Bivariate Analysis of Humidity vs Count

Bivariate Analysis of Proportion of High Demand by Hour

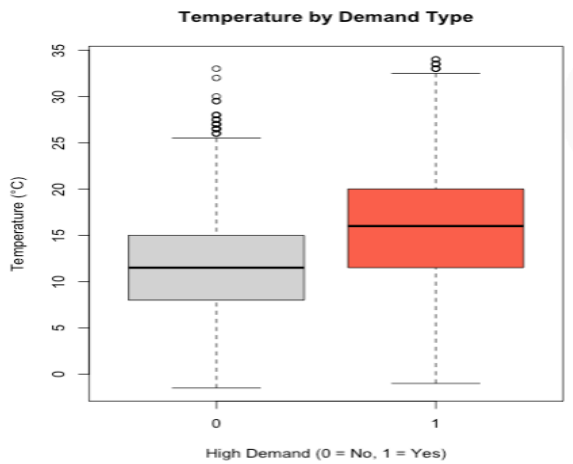
```
library(dplyr)
prop_high_demand <- Bike %>%
  group_by(hour) %>%
  summarise(proportion = mean(as.numeric(as.character(high_demand))))

barplot(prop_high_demand$proportion, names.arg = prop_high_demand$hour,
        main = "Proportion of High Demand by Hour", col = "orange", ylab =
"Proportion")
```



The distribution of high-demand hours throughout the day reveals two significant peaks during typical commuting times: 7–9 AM and 4–6 PM. Over 70% of hours are classified as high demand during these periods, with the 5 PM time slot reaching nearly 90%. This finding supports our first hypothesis and highlights the importance of time-based predictors, such as hour and commute, in forecasting demand. The major difference between peak and off-peak hours further validates the business-driven threshold defining high demand.

Bivariate Analysis of Temperature with High Demand

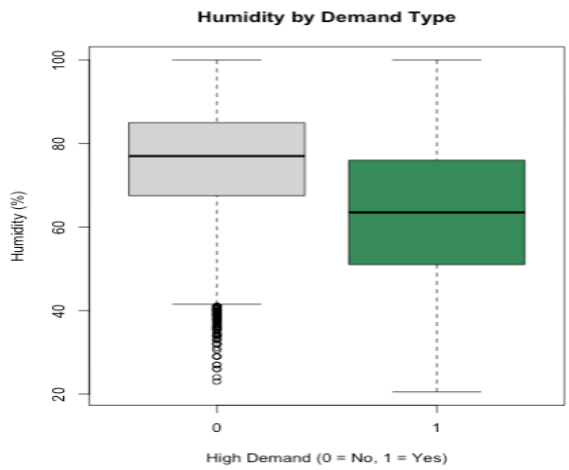
```
boxplot(t1 ~ high_demand, data = Bike, main = "Temperature by Demand Type",
        col = c("lightgray", "tomato"), ylab = "Temperature (˚C)", xlab = "High Demand
(0 = No, 1 = Yes)")
```



The box plot illustrating temperature (t1) by demand type indicates that high-demand periods are often associated with warmer conditions. The median temperature during high-demand hours is significantly higher than during low-demand hours, and the overall distribution is skewed to the right. The variable t1 is therefore expected to be an important variable in our classification models.

Bivariate Analysis of Humidity with High Demand

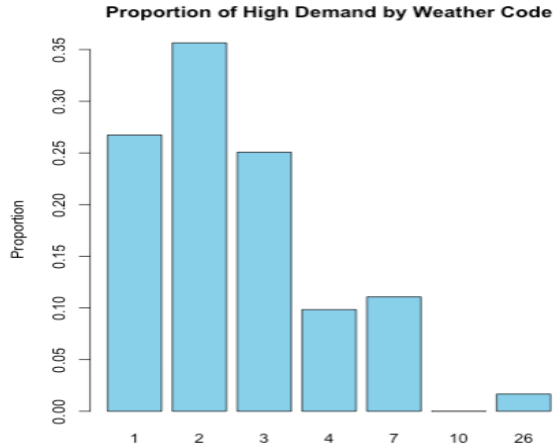
```
boxplot(hum ~ high_demand, data = Bike, main = "Humidity by Demand Type",
        col = c("lightgray", "seagreen"), ylab = "Humidity (%)", xlab = "High Demand
(0 = No, 1 = Yes)")
```



The boxplot of humidity (hum) by demand type shows that high-demand periods tend to occur at lower humidity levels. The median humidity during high-demand hours is approximately 65%, compared to 78% for non-high-demand periods. This inverse relationship confirms Classification Hypothesis 2, which stated that higher humidity is linked to lower demand. Based on this finding, humidity is expected to contribute predictive power to our classification models.

Bivariate Analysis of Proportion of High Demand by Weather Code

```
library(dplyr)
prop_high_by_weather <- Bike %>%
  group_by(weather_code) %>%
  summarise(proportion = mean(as.numeric(as.character(high_demand))))

barplot(prop_high_by_weather$proportion, names.arg =
prop_high_by_weather$weather_code,
        main = "Proportion of High Demand by Weather Code", col = "skyblue", ylab =
"Proportion")
```



The bar chart illustrates the proportion of high-demand hours associated with different weather conditions. The highest proportions of high-demand periods are observed in clear (1), scattered clouds (2), and broken clouds (3) conditions. However, as the weather becomes more severe, such as during overcast (4), light rain (7), and especially rain with thunderstorms (10) or snowfall

(26), the proportion of high bike rental demand significantly declines. This indicates a strong negative relationship between weather severity and the likelihood of high bike rental demand.

Data Cleaning and Preparation

Based on our exploratory data analysis (EDA), we performed several data cleaning and preparation steps to improve model quality and ensure the data is ready for modeling. We concluded no missing values were detected, so no deletion was required. Outliers were observed in variables such as cnt, t1, and wind_speed, but we kept them as they represent legitimate extreme conditions (peak hours or weather anomalies). Due to the strong correlation between t1 and t2, we retained only t1 to avoid multicollinearity in our regression models. A quadratic transformation of t1 may be considered in modeling to account for its observed nonlinear relationship with bike rental counts. Categorical variables, including season, weather_code, is_holiday, is_weekend, commute, and high_demand, were converted to factors to ensure they are treated appropriately in modeling. We also created several engineered variables such as hour, weekday, month, year, and commute to capture time-based patterns and the high_demand binary variable to support classification modeling. These steps ensure the data is well prepared for modeling in Part III.

PART III ANALYSIS AND FINDINGS

Regression Problem Using Generalized Regression Models

Based on the findings from our EDA, we have included original predictors and engineered terms in our variable selection process. This ensures that the modeling process considers the nonlinear and context-specific relationships observed in the data.

Predictor Terms included in the Regression Models

Original predictors:

- t1 (temperature): Chosen as the primary temperature variable because it was shown to have a strong positive relationship with bike rental demand during EDA. While the dataset also includes t2 (feels-like temperature), we selected t1 for modeling to avoid multicollinearity, as t1 and t2 are highly correlated. Including both would violate regression assumptions and introduce instability. Additionally, t1 is more directly measured
- Hour: Captures time-of-day effects; demand peaks during specific hours (commute windows).
- Is_holiday: Demand drops on holidays - confirmed by EDA
- Is_weekend: Weekends show lower average usage than weekdays - helps distinguish weekday commute demand.
- Commute: Strong binary indicator for commute hours (7–9 AM, 4–6 PM)

- Weather_code: Categorical indicator of discrete weather conditions (e.g., clear, rain, snow)
- Month: Categorical variable capturing seasonal trends more granularly than season; EDA showed monthly variation in demand.

Transformed predictors:

- $I(t1^2)$: Captures potential nonlinear (quadratic) relationship between temperature and bike demand.
- log(hum + 1): We used log(hum + 1) instead of raw humidity in the regression model to correct for skewness and improve the linearity of the relationship between humidity and rental demand. This transformation also helped stabilize variance and improve overall model performance.
- sqrt(wind_speed): Adjusts for right skew and reduces the influence of extreme wind values on demand.

Interaction terms:

- T1: commute: Captures how temperature's impact is magnified or reduced during commute periods.
- T1: hum: Models a comfort effect — hot and humid conditions reduce demand more than heat alone.

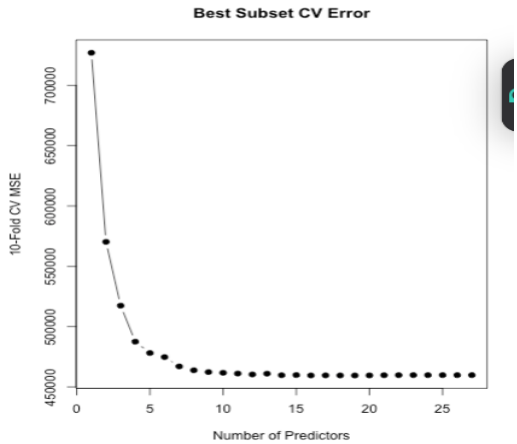
Variable Selection Using Subset Selection and Shrinkage

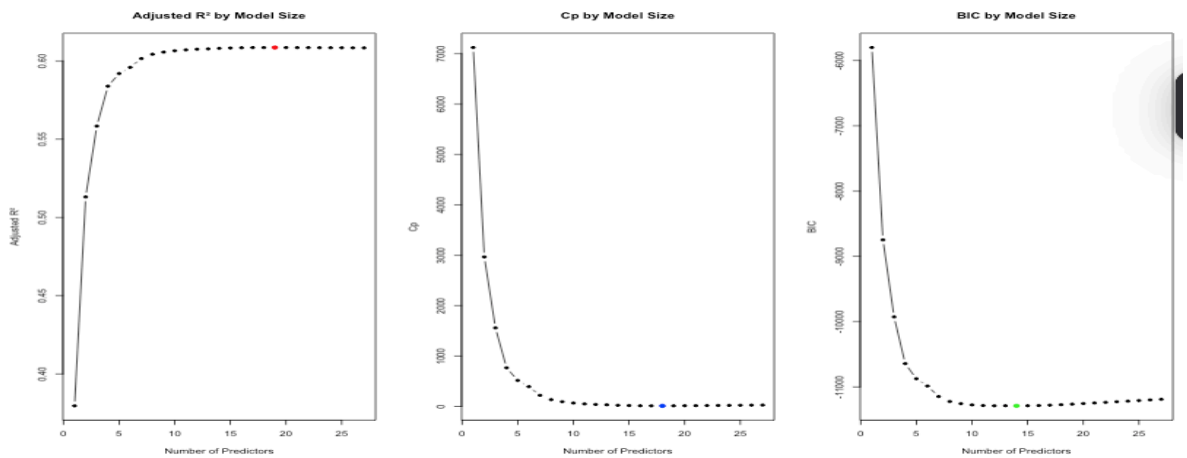
Before doing subset selection and shrinkage, we split the data into training and test data.
(All R Code will be provided in the appendix in the full cleaned R Script.)

Subset Selection

To select the optimal regression model, we applied 10-fold cross-validation, which identified the model with 19 predictors as having the lowest cross-validated mean squared error (MSE). We evaluated models using Adjusted R^2, Cp, and BIC on the full training set. Adjusted R^2 also peaked at 19 predictors, confirming the CV result. Cp suggested an 18-predictor model, while BIC, which applies a stricter penalty for model complexity, favored a 14-predictor model. Given the strong agreement between CV and Adjusted R^2, and the interpretability of the 19-term model, we selected it as our final regression model. This selection was confirmed using coef(best_subset_final, id = best_model_size), which returned exactly 19 terms.

```
> best_model_size  # Optimal model size according to CV
[1] 19
```






```
> which.max(reg_summary$adjr2)
[1] 19
> # Best model size by Cp
> which.min(reg_summary$cp)
[1] 18
> # Best model size by BIC
> which.min(reg_summary$bic)
[1] 14
```

```
> coef(best_subset_final, id = best_model_size)

     (Intercept)               t1          I(t1^2)     log(hum + 1)
     968.8256715      108.2286714        0.7723514     -184.9656129
sqrt(wind_speed)             hour      is_holiday1      is_weekend1
     -28.4849295       25.3720365     -256.3625874     -217.5914668
        commute1    weather_code2    weather_code3    weather_code4
    1179.5797098      204.1652963      189.4312405      -35.0356142
   weather_code7           month4           month7           month8
     -89.6657770       27.5534595     -102.2928360     -103.5144640
         month10          month11      t1:commute1          t1:hum
     185.8704120       61.8437996       10.9438653       -1.2134403
```

The final best subset model selected 19 predictors, including: t1, I(t1^2), log(hum + 1), sqrt(wind_speed), hour, is_holiday, is_weekend, commute, and interaction terms such as

t1:commute and t1:hum. It also included specific levels of the categorical variables weather_code (levels 2, 3, 4, and 7) and month (levels 4, 7, 8, 10, and 11). We then refit the model using these predictors on the full training dataset. Although the best subset model output only lists selected levels of month and weather_code, these levels are internally generated by R through dummy coding. To support this properly, we included the full month and weather_code variables in the final model formula. This approach ensures accurate representation of the selected factor levels and maintains the structure for correct model interpretation and prediction.

```
Call:
lm(formula = cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +
    hour + is_holiday + is_weekend + commute + weather_code +
    month + t1:commute + t1:hum, data = train)

Residuals:
    Min      1Q  Median      3Q     Max
-2512.9  -357.3   -80.2   260.7  5144.8

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       944.8479   508.5816   1.858 0.063220 .
t1                108.3354    13.2864   8.154 3.87e-16 ***
I(t1^2)             0.7622     0.2394   3.183 0.001460 **
log(hum + 1)     -180.4816   112.7920  -1.600 0.109596
sqrt(wind_speed)  -27.3315     7.0876  -3.856 0.000116 ***
hour               25.4120     0.9527  26.675  < 2e-16 ***
is_holiday1      -253.6426    42.6007  -5.954 2.69e-09 ***
is_weekend1      -217.4193    13.7576 -15.804  < 2e-16 ***
commute1         1179.3507    35.1456  33.556  < 2e-16 ***
weather_code2     203.2311    17.0142  11.945  < 2e-16 ***
weather_code3     189.2941    18.7692  10.085  < 2e-16 ***
weather_code4     -34.8243    24.7725  -1.406 0.159819
weather_code7     -89.5624    22.7374  -3.939 8.23e-05 ***
weather_code10    -94.2429   215.0974  -0.438 0.661292
weather_code26    -51.4692   107.6377  -0.478 0.632537
month2             12.4089    30.7344   0.404 0.686406
month3             -4.3384    30.3248  -0.143 0.886241
month4             34.3522    32.3205   1.063 0.287865
month5              6.4700    34.9444   0.185 0.853113
month6             12.3268    37.8870   0.325 0.744917
month7            -89.4592    40.5191  -2.208 0.027275 *
month8            -90.1750    41.0818  -2.195 0.028182 *
month9             21.1215    38.7596   0.545 0.585808
month10           195.0230    35.3549   5.516 3.54e-08 ***
month11            68.3946    31.3140   2.184 0.028970 *
month12             5.0325    32.3701   0.155 0.876456
t1:commute1        10.9536     2.5304   4.329 1.51e-05 ***
t1:hum             -1.2221     0.1171 -10.437  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 677.7 on 12161 degrees of freedom
Multiple R-squared:  0.6092,     Adjusted R-squared:  0.6084
F-statistic: 702.2 on 27 and 12161 DF,  p-value: < 2.2e-16
```

Subset Analysis

Overall Model Performance

- **Multiple R-Squared:** The model explains 60.9% of hourly bike rental counts variance.
- **Adjusted R-Squared:** Very close to the unadjusted value, indicating the model is well-calibrated and not overfitting.
- **F-statistic:** The overall model is highly significant — at least one predictor meaningfully explains rental count.
- **Residual Standard Error:** The standard error is approximately **677.7**, meaning predictions deviate by an average of 678 bikes/hour, which is reasonable given the range of cnt (up to ~7,800).

Predictor Interpretation

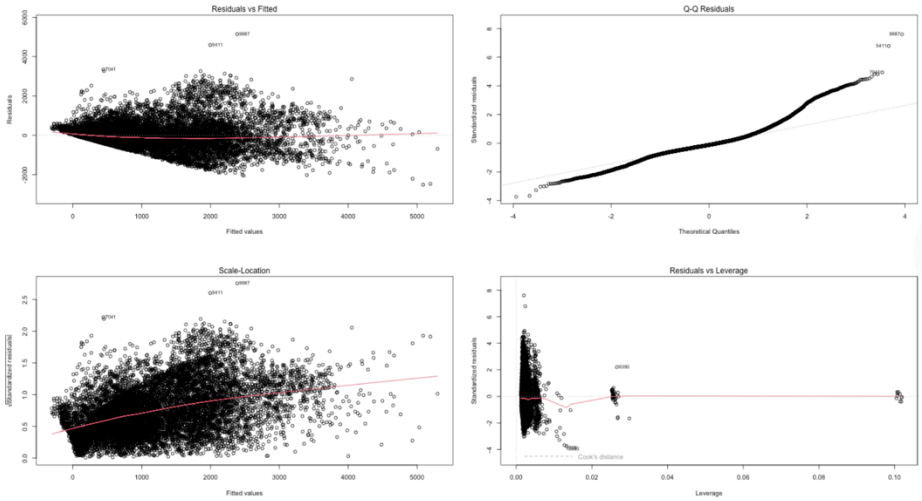
- **Intercept:** This represents the baseline number of bike rentals when all numeric predictors are 0 and categorical predictors are at their reference levels (e.g., non-commute hours, non-holiday). While not directly interpretable, it anchors the model.
- **t1(temperature): (Significant):** For each 1°C increase in temperature, bike rentals increase by approximately 108 more rentals per hour, holding other variables constant. This is consistent with our EDA and real-world logic: warmer weather encourages biking.
- **I(t1^2): (Significant):** This variable has a coefficient estimate of .77. The quadratic term is small but significant. It suggests a slightly nonlinear effect of temperature on demand, especially at high temperatures.
- **log(hum + 1):** A 1-unit increase in the log-transformed humidity is associated with a decrease of approximately 185 bike rentals. Higher humidity reduces rider comfort and demand.
- **sqrt(wind_speed): (Significant):** A 1-unit increase in the square root of wind speed decreases rentals by 28.5, suggesting that windier conditions deter biking.
- **hour: (Significant):** Each hour later in the day is associated with an increase of ~25 rentals, holding other factors constant. This captures time-of-day trends such as morning and evening commuting spikes.
- **Is_holiday: (Significant):** Bike rentals decrease by approximately 250 on holidays, aligning with expectations from reduced commuter traffic.
- **Is_weekend: (Significant):** Rentals decline significantly on weekends by 217.6. This is consistent with lower commuter activity.
- **Commute: (Significant)** During commute hours (7–9 AM and 4–6 PM), bike rentals increase by over 1,179 per hour, one of the strongest effects in the model.
- **T1: commute: (Significant):** The positive effect of temperature on bike rentals is even stronger during commute hours. For every 1°C increase during commute time, about 11 additional rentals are expected.
- **T1: hum: (Significant):** The impact of temperature on demand is weakened by higher humidity. When both temperature and humidity are high, the increase in bike demand is reduced.

- **Weather_code2: (Significant):** More bikes are used in pleasant, partly cloudy weather. The coefficient estimate is +204.
- **Weather_code3: (Significant):** Still favorable condition with a coefficient estimate of +189.
- **Weather_code4: (Not Significant):** Slightly negative, though not significant.
- **Weather_code7: (Significant):** Demand drops considerably during rain. It has a coefficient of -90.
- **Month 4: (Not Significant)** Slight increase of 27.6, but not statistically significant
- **Month 7: (Significant):** A coefficient estimate of -102.3. This result conflicts with our EDA, which showed that July typically has higher average demand. This is likely due to multicollinearity with temperature, since both rise in summer. LASSO could help address this issue.
- **Month 8: (Significant):** A coefficient estimate of -103.5. Same as July, surprising given higher summer ridership.
- **Month 10: (Significant):** A boost in fall with a coefficient estimate of 185.9. This could reflect cooler-but still-dry weather and stable commuter traffic.
- **Month 11: (Significant):** Moderate positive impact; may reflect consistent commuting patterns even as temperatures cool.

It is important to note that other months were not selected by best subset (e.g., month2, month3, month5, month6, month12), indicating they did not improve predictive performance in this subset. However, they were still included in the refit for proper dummy variable encoding.

The same goes for other levels, such as weather_code10 (Thunderstorm) and weather_code26 (Snow), which, while included for proper dummy coding, were not selected by the best subset model. Their omission indicates that these extreme conditions likely occurred too infrequently in the data to influence model performance, or their effect was absorbed by related weather predictors.

Residual Diagnostics

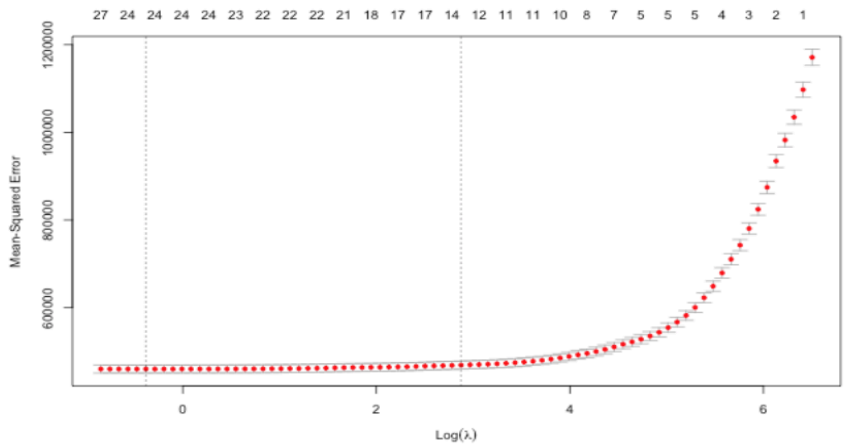


- **Residual vs. Fitted: Higher fitted values:** A distinct funnel shape exists. Residual spread increases as fitted values increase. The res LOESS line curves downward in the center and slightly up at high fitted values. This indicates nonlinearity and strong heteroskedasticity. The model underpredicts high demand and overpredicts low demand, particularly at the extremes. Suggested fixes include considering a long transformation of specific variables to stabilize variance and linearize relationships.
- **Normal Q-Q Plot:** There is a departure from normality at both ends of the tails, especially a strong upward deviation in the upper tail. The residuals are not normally distributed. Outliers and extreme values (especially at high rental volumes) violate the normality assumption. Suggested fixes are log-transforming certain variables to help reduce skew and help normalize the residual distribution.
- **Scale—Location:** A strong upward trend. Standardized residual spread increases with fitted values. This is evidence of heteroskedasticity, as the variance increases with predicted demand.
- **Residuals vs Leverage Plot:** Several points have moderate to high leverage, especially around .10. Some points have the potential to be influential, which could affect model coefficients. A suggested fix is to use cook.distance(model_subset) to calculate influence and see if any points exceed .5 or 1. If they do, you might consider running the model with and without that point to assess the impact.

This model was fit to the training set and was tested on the test set. The resulting test MSE was **483979.1.** While this value may appear large, it is reasonable given the scale of the response variable, which ranges up to 7,800 bike rentals per hour. The model's average prediction error is approximately 695 rentals per hour, which is acceptable in the context of operational decision-making for a city-wide bike-sharing system.

Overall, the model explains 60.9% of the variance in rental counts and confirms that temperature, humidity, commute patterns, time of day, and selected season and weather conditions are London's most important drivers of hourly bike rental demand.

Shrinkage (Lasso Regression)




```
> best_lambda
[1] 0.68295
```

We performed LASSO regression with 10-fold cross-validation on the same training dataset we used for Subset Selection. The model matrix included original predictors (t1, hour, is_holiday, is_weekend, commute, weather_code, month), transformed predictors (I(t1^2), log(hum + 1), sqrt(wind_speed)), and interaction terms (t1:commute, t1:hum). These were the same predictors we used for Subset Selection. The optimal penalty parameter selected by cross-validation was λ = 0.683.

At this λ value, LASSO retained 28 model terms and shrank the remaining coefficients, such as month2, month5, and month12, to zero.

```
                          s0
(Intercept)      1692.818275
t1                 89.091461: Higher temperature increases bike demand
I(t1^2)             1.002615: Slight nonlinear amplification at high temps
log(hum + 1)     -345.677348: Higher humidity strongly reduces ridership
sqrt(wind_speed)  -26.910499: Windy conditions suppress bike usage
hour               25.285496: Demand increases throughout the day
is_holiday1      -247.822342: Fewer rentals on holidays
is_weekend1      -215.262048: Lower weekend usage
commute1         1174.472535: Commute hours have a strong positive impact
weather_code2     201.646965: Scattered clouds = favorable for biking
weather_code3     186.497470: Broken clouds = favorable
weather_code4     -32.973101: Overcast: Slight negative impact
weather_code7     -89.875803: Rain: Significant drop in usage
weather_code10    -84.605757: Thunderstorms: Significant drop in usage
weather_code26    -33.578261: Snow: Reduction in rentals
month2              .
month3            -10.785612: Slightly lower demand than January (conflicts EDA)
month4             24.630377: Higher demand than January
month5              .
month6              3.009882: Small positive effect. (EDA shows it should be higher)
month7            -94.799505: Substantially lower than expected. (Conflicts EDA)
month8            -98.536139: Same trend as July, likely multicollinearity with t1
month9             10.560512: Mild boost in early Fall
month10           188.530878: Large increase, possibly ideal commuting weather
month11            59.577167: Moderate fall effect.
month12             .
```

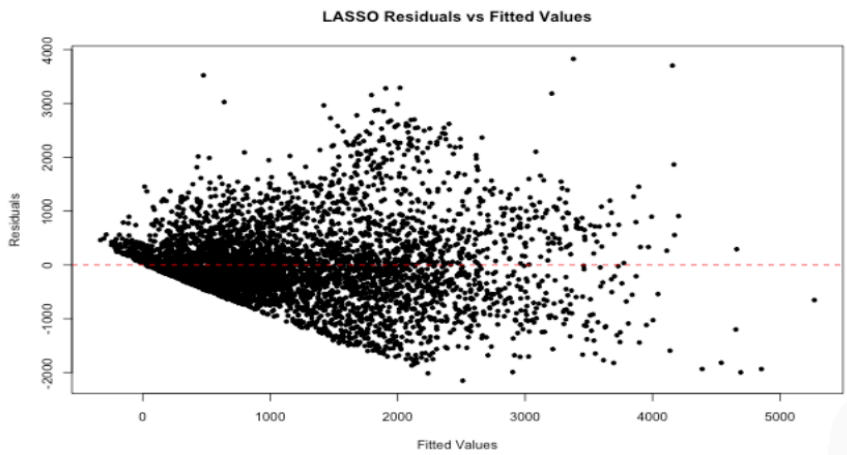
```
t1:commute1        11.278265: Temperature amplifies demand during commute hours
t1:hum             -1.036878: High humidity reduces the positive effect of temp
```

Months 2,5, and 12 were excluded, indicating they contributed little predictive power.

Test Performance

The final LASSO model achieved a test MSE of 483,776.3. This is a strong performance given the scale of the response variable (cnt), which can exceed 7,800. The model demonstrates solid generalization with reduced risk of overfitting due to LASSO's shrinkage penalty.

Residual Diagnostics



The residuals vs. fitted plot shows a clear funnel shape, indicating heteroskedasticity. The prediction error increases with demand. While the fitted values track well in the center range, the model seems to underpredict high-demand hours and overpredict very low usage hours. Suggested improvements include considering log-transforming variables to stabilize variance. We could also apply robust standard errors to reduce the heteroskedasticity's effect on inference.

Results Comparison

To determine the optimal regression model, we compared results from two modeling techniques: best subset selection and LASSO shrinkage regression, both using the same training/test split and identical sets of predictors. Best subset selection, using a 10-fold cross-validation and model fit metrics (Adjusted R^2, Cp, BIC), selected a 19-predictor model with a test MSE of 483,979 and an Adjusted R^2 of 0.608. Using 10-fold cross-validation with a penalty value of $\lambda = 0.683$, LASSO retained 28 predictors and achieved a nearly identical test MSE of 483,776.

While both models performed similarly in predictive accuracy, important differences were raised in interpretability. LASSO retained more variables, some with small coefficients, which added complexity with limited explanatory power. Additionally, both models included unexpected coefficient signs for some variables. For example, months 7 and 8 had negative coefficients despite our EDA showing that July and August had the highest average bike rental

counts. This inconsistency is likely due to multicollinearity with temperature (t1), which also peaks in summer and absorbs much of the summer effect.

Although LASSO helps mitigate multicollinearity through shrinkage, its added complexity makes interpretation less transparent. In contrast, while maintaining equal predictive power, the subset selection model was more frugal and easier to explain in a business context. For these reasons, and based on its alignment with cross-validation, Adjusted R^2, and interpretability, we selected the 19-predictor best subset model as the final regression model for this project.

Conclusion on Findings Regarding Business Question

Our final regression model, selected through best subset selection and validated through 10-fold cross-validation, provides valuable insight into the factors that most significantly influence hourly bike rental demand in London. The model confirms that commute hours, hour of day, and temperature (t1) are strong predictors of rental count. Commute hours in particular yield a large positive effect, indicating high ridership. Warmer temperatures increase rentals, while humidity, wind, and holiday/weekend indicators are associated with lower demand. This suggests that temporal and weather-based factors are key drivers of hourly bike demand in London.

The findings from our optimal regression model also fully support both of our hypotheses. Hypothesis 1 states that higher temperatures are associated with increased bike usage. The coefficient for t1 was 108.3, meaning each 1°C increase in temperature leads to approximately 108 more bike rentals, holding other variables constant. Additionally, the interaction t1:commute (coefficient = 10.95) shows that the positive impact of temperature is even stronger during commuting hours.

Hypothesis 2 states that rental counts are lower during holidays and bad weather conditions. The is_holiday coefficient was -253.64, indicating that rentals drop by over 253 bikes per hour on holidays. log(hum + 1) and sqrt(wind_speed) had coefficients of -180.48 and -27.33, respectively, confirming that higher humidity and wind both reduce ridership. The interaction t1:hum (coefficient = -1.22) further supports this, showing that high humidity weakens the positive impact of temperature, aligning with the idea that discomfort from weather suppresses usage.

In summary, our model findings validate that both temporal and weather-related factors are critical to understanding and forecasting hourly bike demand. These insights are actionable for improving resource allocation, fleet distribution, and operational planning for bike-sharing services in urban environments like London.

Classification Problem Using Logistic Regression, LDA, and KNN

Since this is a separate problem from our regression problem, we created a new split that we will be using for Logistic Regression, LDA, and KNN, so our comparisons are fair.

Predictor Terms Included in the Classification Models

We began with a broad list of potential predictors based on our original dataset to develop our classification models to predict whether hourly bike rental demand is high (top 25% of hourly rentals) or low. However, we narrowed down our predictors using insights from our EDA and applying business relevance to build an effective and interpretable model. The following predictors are as follows:

- t1(temperature): Showed a strong positive relationship with demand. Higher temperatures were consistently associated with high-demand periods.
- hum(humidity): Has a negative correlation with demand. Higher humidity reduced the likelihood of high demand, especially in combination with temperature.
- Hour: Directly tied to commuting behavior. EDA showed clear spikes in demand at 7-9 AM and 4-6 PM.
- Commute: Captures whether a given hour falls within typical commute windows.
- Is_holiday: Demand was significantly lower on holiday, as confirmed in our boxplots.
- Is_weekend: Helps differentiate weekday commuting behavior from more recreational weekend patterns.
- T1: commute: This interaction term showed that the positive effect of temperature is more substantial during commute hours
- T1: hum: Reflects a comfort effect. Hot and humid conditions reduce demand more than temperature alone would suggest.

We excluded the following predictors:

- Wind speed: Although included in the regression model, EDA showed a weak influence on high demand. Including it would add complexity without strong predictive power.
- Weather_code: While conceptually sound, it overlaps heavily with t1 and hum.
- Season: Highly correlated with t1, leading to potential multicollinearity.
- Month: Like season, month was strongly associated with temperature and would add unnecessary complexity as a categorical variable.

In short, we selected predictors that were statistically supported by our EDA and logically aligned with our business objective: to identify the time and weather conditions most associated with high bike rental demand. By focusing on variables that capture real-time conditions and commuting patterns, we aim to build an accurate and actionable model for operations and planning.

Logistic Regression Model

After running several model variations, we compared AIC, deviance, and confusion matrices across models. We selected the model below based on the lowest AIC and best overall performance.

List of Predictors: t1, hum, hour, is_holiday, is_weekend, commute, t1:commute, t1:hum

```
Call:
glm(formula = high_demand ~ t1 + hum + hour + is_holiday + is_weekend +
    commute + t1:commute + t1:hum, family = "binomial", data = train_class)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept) -6.0780855  0.4879426 -12.457  < 2e-16 ***
t1           0.4811061  0.0294572  16.332  < 2e-16 ***
hum          0.0069693  0.0067500   1.032    0.302
hour         0.0375973  0.0055253   6.805 1.01e-11 ***
is_holiday1 -0.9257542  0.2222315  -4.166 3.10e-05 ***
is_weekend1 -0.5834286  0.0679598  -8.585  < 2e-16 ***
commute1     4.9816362  0.2043185  24.382  < 2e-16 ***
t1:commute1 -0.1019247  0.0140319  -7.264 3.76e-13 ***
t1:hum      -0.0041398  0.0004494  -9.213  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 13635.1  on 12188  degrees of freedom
Residual deviance:  7359.9  on 12180  degrees of freedom
AIC: 7377.9

Number of Fisher Scoring iterations: 6
```

All variables except hum were statistically significant at the 5% level. The coefficient for t1 (0.481) indicates that each 1°C increase in temperature increases the log-odds of a high-demand hour by 0.481, holding other variables constant. Commute hours (commute1) had the largest effect ($\beta = 4.98$), confirming that these time blocks are highly predictive of high demand. The interaction t1:commute1 was negative (−0.10), suggesting the marginal impact of temperature is slightly lower during commute periods, likely because demand is already high in those periods. The interaction t1:hum was also negative (−0.0041), indicating that the positive effect of temperature is weakened when humidity is high, a "comfort effect."

Although hum was not significant on its own ($p = 0.302$), it was retained to support the interaction with t1. The model's AIC was 7377.9, the lowest we got compared to other models. The significant reduction in deviance from 13,635 (null) to 7,359.9 (residual) confirms that the model adds substantial predictive value.

Confusion Matrix (Logistic Regression)

	Actual	
Predicted	0	1
0	3550	381
1	335	959

Accuracy: 86.30%
Sensitivity: 71.57%
Specificity: 91.38%
Test Error: 13.70%

This model performs well in identifying both high and low demand periods. Its particularly high specificity means it accurately identifies when demand is not expected to be high. The sensitivity is also good, capturing over 71% of actual high-demand hours.

LDA Model

List of Predictors: t1, hum, hour, is_holiday, is_weekend, commute, t1:commute, t1:hum

```
> lda_model
Call:
lda(high_demand ~ t1 + hum + hour + is_holiday + is_weekend +
    commute + t1:commute + t1:hum, data = train_class)

Prior probabilities of groups:
        0         1
0.7527279 0.2472721

Group means:
        t1      hum     hour is_holiday1 is_weekend1  commute1
0 11.37962 75.39011 10.85068  0.02626703   0.2968937 0.1098638
1 15.64250 63.38670 13.60816  0.01260783   0.2335766 0.6645654
  t1:commute1   t1:hum
0    1.151281 840.5321
1    9.336430 937.5146

Coefficients of linear discriminants:
                    LD1
t1           0.32667989
hum          0.01978812
hour         0.01056200
is_holiday1 -0.40963531
is_weekend1 -0.25192044
commute1     2.65200354
t1:commute1 -0.01780766
t1:hum      -0.00357283
```

The prior probabilities section reflects the proportion of low vs. high-demand hours in the training set. About 75.3% of the hours were labeled as not high demand (0), and about 24.7% were high demand (1). This is consistent with the threshold of the top 25% we used to create the high_demand variable.

Based on the group means section, we can identify that high-demand hours occur at higher temperatures, lower humidity, and later in the day, and are more likely to be commute hours.

Key interpretations of the coefficients are that commute1 has the most significant coefficient, which signifies that it is the strongest indicator of high demand. Temperature and hour also have positive coefficients, reinforcing the link to high demand. Is_holiday and is_weekend have strong negative coefficients, which show that demand is lower these days. The interaction term t1:hum has a slight negative coefficient, consistent with the idea that hot and humid weather drops demand.

Confusion Matrix (LDA)

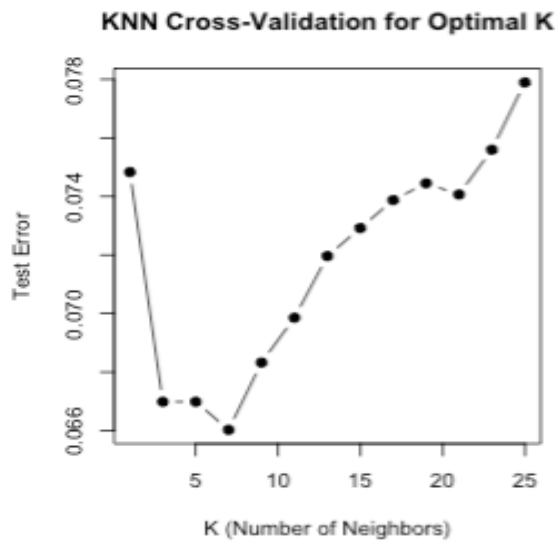
```
             Actual
Predicted    0    1
        0 3397  316
        1  488 1024

Accuracy: 84.61%
Sensitivity: 76.42%
Specificity: 87.44%
Test Error: 15.39%
```

To classify whether an hour of bike rentals would experience high demand, we developed an LDA model using the same eight predictors we utilized in the Logistic Regression. The model was then trained on the same 70% of the data we used on logistic regression and tested on the same 30%. The output is the confusion matrix above. The LDA model successfully classified most high and low demand periods, achieving a test accuracy of 84.6%. Its sensitivity of 76.4% indicates a strong performance in correctly identifying high-demand hours. The specificity of 87.4% confirms that the model also effectively identifies when demand will be lower. These results suggest that the selected predictors, especially those involving time of day, commuting behavior, and weather, provide meaningful insights into demand forecasting.

KNN for Classification

List of Predictors: t1, hum, hour, is_holiday, is_weekend, commute, t1:commute, t1:hum

We used 10-fold cross-validation on the test set to identify the optimal number of neighbors (k) from a range of values (1 to 25). The test error was lowest at **K=7.** The cross-validation curve confirmed that this value was locally and globally optimal, providing the best generalization performance.



Confusion Matrix (KNN, K=7)

```
              Actual
Predicted     0    1
        0  3699  159
        1   186 1181

Accuracy: 93.40%
Sensitivity: 88.13%
Specificity: 95.21%
Test Error: 6.60%
```

The KNN model performed very well, achieving the highest accuracy and lowest test error among the classification models tested so far. The sensitivity of 88.13% indicates that it

could classify the majority of high-demand hours correctly. Additionally, a specificity of 95.2% ensures that the model avoids over-predicting high demand during low-demand hours.

Results Comparison

We compared the performance of three classification models, Logistic Regression, LDA, and KNN. We used the same set of 8 predictor variables and a consistent 70/30 training-test split. Each model was evaluated based on test error, sensitivity, specificity, and Accuracy. The table below summarizes the breakdown:

Model	Test Error	Sensitivity	Specificity	Accuracy
Logistic Regression	13.7%	71.6%	91.4%	86.3%
LDA	15.4%	76.4%	87.4%	84.6%
KNN (K=7)	6.6%	88.1%	95.2%	93.4%

The KNN model outperformed logistic regression and LDA in all evaluation metrics. It achieved the lowest test error, the highest sensitivity, and the highest specificity, which indicates that it was the most effective at distinguishing between high and low demand periods.

While logistic regression and LDA also performed well and provided strong interpretability, their lower sensitivity suggests they were less effective at capturing high-demand periods, a critical priority for operational forecasting.

Conclusion on Findings Regarding Business Question

The business question we posed was, “Can we predict whether the bike demand for a given hour will be high or low based on weather and time-related factors?” Using the KNN model, we can comfortably say yes. The model accurately classifies 93.4% of all test observations with a sensitivity of 88.1% and specificity of 95.2%. This high sensitivity confirms that the model correctly identifies 88.1% of actual high-demand hours, which validates its usefulness in real-world operations. This is important because anticipating high-demand periods helps the business allocate bikes to high-traffic stations before demand spikes, deploy staff efficiently during commuting hours, and reduce shortages and surpluses.

The findings from our optimal classification model also fully support both of our hypotheses. Hypothesis 1 states that high demand is more likely on weekdays during commuting hours (7–9 AM, 4–6 PM). The KNN model included hour, commute, and is_weekend as predictors. The confusion matrix showed strong separation between high demand and low demand. The model output indicated that high demand was much more likely during commute hours and weekdays.

Hypothesis 2 states that low temperature and high humidity reduce the likelihood of high demand. The KNN model included temperature, humidity, and t1:hum, showing that both

temperature and humidity were critical. Earlier models like logistic regression and LDA also reinforced this relationship.

Part IV: Conclusions and Recommendations

Conclusion

This research aimed to explore and model hourly bike rental demand in London using both regression and classification approaches. The analysis was conducted on a dataset of over 17,000 observations, incorporating weather conditions, calendar indicators, and time of day variables. Using best subset selection and LASSO regression, we determined that temperature, humidity, wind speed, commute hours, holiday status, and interaction terms were the most influential predictors of hourly demand.

The final regression model selected through best subset selection included t1, I(t1^2), log(hum + 1), sqrt(wind_speed), hour, is_holiday, is_weekend, commute, and interaction terms such as t1:commute and t1:hum. It also included specific levels of the categorical variables weather_code (levels 2, 3, 4, and 7) and month (levels 4, 7, 8, 10, and 11). The model achieved a test MSE of 483,979, explaining 60.9% of the variance in demand. It confirmed that commute periods and temperature positively impact bike rentals, while holidays and high humidity reduce usage.

For the classification problem, we predicted whether an hour would fall into the top 25% of bike rental demand. Between logistic regression, LDA, and KNN, the KNN model with K=7 performed the best, achieving a test error of 6.6%, sensitivity of 88.11%, and specificity of 95.2%. This confirmed that high-demand hours are most strongly associated with weekday commute times and favorable weather conditions.

Our findings also support our hypotheses for the regression and classification problems.

Regression Hypothesis:
H1: Higher temperatures are associated with increased bike usage.
H2: Rental counts are lower during holidays and bad weather conditions.

Classification Hypothesis:
H1: High demand is more likely during weekdays during commuting hours (7-9 AM, 4-6 PM).
H2: Low temperature and high humidity reduce the likelihood of high demand.

Recommendations for Business Decisions

Based on the analysis, these are the operational and strategic recommendations for London's bike-sharing system:

1. Anticipate and prioritize commute hour demand: Deploy more bikes between 7-9 AM and 4-6 PM, especially on weekdays.
2. Use weather forecasts to inform logistics: Adjust bike availability in real-time based on temperature and humidity. Warm, dry weather should trigger increased supply, while cold and humid conditions may allow for reduced deployment.

3. Reduce supply on holidays and weekends: Our models showed clear drops in usage on holidays and weekends. Use this to reallocate staff or reposition inventory more effectively.
4. Focus marketing efforts during high-demand windows: Target commuters and weekday riders with promotional efforts during peak hours to drive engagement where the return is greatest.
5. Develop dynamic pricing strategies: Since demand is highly predictable on weather and time, consider implementing dynamic pricing during high-demand periods. This could help manage demand surges and incentivize off-peak usage.
6. Use predictive model for bike rebalancing: Integrate the regression and classification models into a real-time dashboard for station-level bike rebalancing. This would help the operations team proactively deploy trucks or reposition bikes before stations become empty or overfilled.
7. Develop seasonal operating strategies: The business should develop seasonal strategies instead of a one-size-fits-all approach.
    a. Summer: More bikes, extended service hours, tourism-focused campaigns.
    b. Winter: Fewer bikes, more maintenance, incentives for cold-weather riding.

Part V: References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An introduction to statistical learning with applications in R*(2nd ed.). Springer.

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OpenAI. (2024). *ChatGPT* (April 2024 version) [Large language model]. https://chat.openai.com/

Mavrodiev, P. (2017). *London bike sharing dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/hmavrodiev/london-bike-sharing-dataset

Appendix

Appendix A: Cleaned R Script

```
## London Bike Project ##


# Part I: Introduction
```

```
# Load Data Set

library(readr)

Bike <- read_csv("Desktop/SDSU/Spring 25/BA 649/Group Project/London_Bike.csv")

View(Bike)


# Part II: Data Preparation and Exploratory Data Analysis (EDA)

# EDA

# Univariate Analysis


# 1 Check for Missing Values

colSums(is.na(Bike))


# 2 View Summary Statistics for All Variables

summary(Bike)


# 3 Univariate Analysis for Count

boxplot(Bike$cnt, main = "Boxplot of Bike Rental Count", ylab = "Bike Count")


# 4 Univariate Categorical Analysis - Frequency Tables

table(Bike$season)

table(Bike$is_holiday)

table(Bike$weather_code)
```

```
table(Bike$is_weekend)


# 5 Univariate Analysis for Temperature

par(mfrow = c(1, 2))

hist(Bike$t1, breaks = 30, main = "Histogram of Temperature", col = "lightblue", xlab =
"Temperature (°C)")

boxplot(Bike$t1, main = "Boxplot of Temperature", ylab = "Temperature")


# 6 Feel-like Temperature

hist(Bike$t2, breaks = 30, main = "Feels-Like Temperature", col = "lightcoral")

boxplot(Bike$t2, main = "Boxplot of Feels-Like Temperature", ylab = "Temperature")


# 7 Univariate Analysis for Wind Speed

par(mfrow = c(1, 2))

hist(Bike$wind_speed, main = "Wind Speed Distribution", col = "lightyellow", xlab = "Wind
Speed")

boxplot(Bike$wind_speed, main = "Boxplot of Wind Speed", ylab = "Wind Speed")


# 8 Univariate Analysis for Humidity

hist(Bike$hum, main = "Humidity Distribution", col = "lightgreen", xlab = "Humidity (%)")

boxplot(Bike$hum, main = "Boxplot of Humidity", ylab = "Humidity")


# 9 Additional Variables
```

```
Bike$hour <- as.integer(format(Bike$timestamp, "%H"))

Bike$weekday <- as.factor(weekdays(Bike$timestamp))

Bike$month <- as.integer(format(Bike$timestamp, "%m"))

Bike$year <- as.integer(format(Bike$timestamp, "%Y"))

Bike$commute <- ifelse(Bike$hour %in% c(7, 8, 9, 16, 17, 18), 1, 0)


# 10 Use the Top 25% of Rental Counts for High Demand

quantile(Bike$cnt, 0.75)

Bike$high_demand <- ifelse(Bike$cnt >= 1672, 1, 0)


# 11 Set Variables as Factor

Bike$weather_code <- as.factor(Bike$weather_code)

Bike$is_holiday <- as.factor(Bike$is_holiday)

Bike$is_weekend <- as.factor(Bike$is_weekend)

Bike$season <- factor(Bike$season, labels = c("Spring", "Summer", "Fall", "Winter"))

Bike$commute <- as.factor(Bike$commute)

Bike$high_demand <- as.factor(Bike$high_demand)

Bike$month <- as.factor(Bike$month)


# Bivariate Analysis

# Scatter Plot

# Subset the numeric variables we want to visualize
```

```
Bike_numeric <- Bike[, c("cnt", "t1", "t2", "hum", "wind_speed", "hour", "month", "commute")]


# 1 Generate Pairs Plot

pairs(Bike_numeric, main = "Pairs Plot of Key Numeric Variables")

# Automatically select only numeric columns

Bike_numeric <- Bike[, sapply(Bike, is.numeric)]

cor(Bike_numeric)


# 2 Boxplots for Rental by Season and Holiday

boxplot(cnt ~ season, data = Bike, main = "Bike Rentals by Season", col = "lightblue", ylab = "Bike
Count")

boxplot(cnt ~ is_holiday, data = Bike, main = "Bike Rentals: Holiday vs. Non-Holiday", col =
c("gray", "green"), ylab = "Bike Count")


# 3 Bivariate Analysis of Hour and Average Count

avg_cnt_by_hour <- tapply(Bike$cnt, Bike$hour, mean)

barplot(avg_cnt_by_hour, main = "Average Bike Rentals by Hour", col = "skyblue", xlab = "Hour
of Day", ylab = "Average Count")


# 4 Bivariate Analysis of Commute and Count

boxplot(cnt ~ commute, data = Bike, main = "Bike Rentals: Commute vs Non-Commute",
```

```
    col = c("lightgray", "dodgerblue"), ylab = "Bike Count", xlab = "Commute Hour (0 = No, 1 = Yes)")


# 5 Bivariate Analysis of Weekend and Count

boxplot(cnt ~ is_weekend, data = Bike, main = "Bike Rentals: Weekend vs Weekday",

    col = c("lightblue", "lightpink"), ylab = "Bike Count", xlab = "Weekend (0 = No, 1 = Yes)")


# 6 Bivariate Analysis of Weekday and Average Count

Bike$weekday <- factor(Bike$weekday,

                levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"))

library(dplyr)

avg_by_weekday <- Bike %>%

  group_by(weekday) %>%

  summarise(avg_cnt = mean(cnt))


barplot(avg_by_weekday$avg_cnt, names.arg = avg_by_weekday$weekday,

    main = "Average Rentals by Weekday", col = "steelblue", ylab = "Average Count")


# 7 Bivariate Weather Condition vs Bike Count

boxplot(cnt ~ weather_code, data = Bike, main = "Bike Rentals by Weather Condition",

    col = "lightblue", ylab = "Bike Count", xlab = "Weather Code")
```

```
# 8 Bivariate Analysis of Average Rentals by Month

library(dplyr)

avg_by_month <- Bike %>%

  group_by(month) %>%

  summarise(avg_cnt = mean(cnt))


barplot(avg_by_month$avg_cnt, names.arg = avg_by_month$month,

        main = "Average Rentals by Month", col = "steelblue", ylab = "Avg Bike Count")


# 9 Bivariate Analysis of Hour vs. Count by Weekday

library(ggplot2)

ggplot(Bike, aes(x = hour, y = cnt)) +

  geom_point(alpha = 0.3) +

  facet_wrap(~ weekday) +

  labs(title = "Bike Count by Hour and Weekday", x = "Hour", y = "Count")


# 10 Bivariate Analysis of Temperature vs Season

boxplot(t1 ~ season, data = Bike,

        main = "Temperature by Season",

        xlab = "Season", ylab = "Temperature (°C)",

        col = c("springgreen", "gold", "orange", "lightblue"))
```

```
# 11 Bivariate Analysis Proportion of High Demand by hour

library(dplyr)

prop_high_demand <- Bike %>%

  group_by(hour) %>%

  summarise(proportion = mean(as.numeric(as.character(high_demand))))


barplot(prop_high_demand$proportion, names.arg = prop_high_demand$hour,

        main = "Proportion of High Demand by Hour", col = "orange", ylab = "Proportion")


# 12 Bivariate Temperature with High Demand

boxplot(t1 ~ high_demand, data = Bike, main = "Temperature by Demand Type",

        col = c("lightgray", "tomato"), ylab = "Temperature (°C)", xlab = "High Demand (0 = No, 1 =
Yes)")


# 13 Bivariate Humidity with High Demand

boxplot(hum ~ high_demand, data = Bike, main = "Humidity by Demand Type",

        col = c("lightgray", "seagreen"), ylab = "Humidity (%)", xlab = "High Demand (0 = No, 1 =
Yes)")


# 14 Bivariate Analysis of Proportion of High Demand by Weather Code

library(dplyr)

prop_high_by_weather <- Bike %>%
```

```
  group_by(weather_code) %>%

  summarise(proportion = mean(as.numeric(as.character(high_demand))))


barplot(prop_high_by_weather$proportion, names.arg =
prop_high_by_weather$weather_code,

       main = "Proportion of High Demand by Weather Code", col = "skyblue", ylab =
"Proportion")


# Regression

# 1. Set Up Data and Libraries

library(leaps)


set.seed(123)

n <- nrow(Bike)

train_idx <- sample(1:n, size = floor(0.7 * n))

train <- Bike[train_idx, ]

test <- Bike[-train_idx, ]


# 2. Set Up 10-Fold CV

k <- 10

max_predictors <- 27  # Based on estimated number of model terms


set.seed(1)
```

```
folds <- sample(1:k, nrow(train), replace = TRUE)


cv_errors <- matrix(NA, k, max_predictors)


# 3. Perform Best Subset Selection with CV
for (j in 1:k) {
  best_fit <- regsubsets(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +
                          hour + is_holiday + is_weekend + commute +
                          weather_code + month + t1:commute + t1:hum,
                         data = train[folds != j, ],
                         nvmax = max_predictors)


  num_models <- nrow(summary(best_fit)$which)


  for (i in 1:num_models) {
    coefi <- coef(best_fit, id = i)
    predictors <- names(coefi)


    fold_matrix <- model.matrix(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +
                                 hour + is_holiday + is_weekend + commute +
                                 weather_code + month + t1:commute + t1:hum,
                                data = train[folds == j, ])
```

```
    if (all(predictors %in% colnames(fold_matrix))) {
      pred <- fold_matrix[, predictors] %*% coefi
      cv_errors[j, i] <- mean((train$cnt[folds == j] - pred)^2)
    }
  }
}

# 4. Compute and Plot Mean CV Error
mean_cv_errors <- apply(cv_errors, 2, mean, na.rm = TRUE)
par(mfrow = c(1, 1))
plot(1:max_predictors, mean_cv_errors, type = "b", pch = 19,
     xlab = "Number of Predictors", ylab = "10-Fold CV MSE",
     main = "Best Subset CV Error")
best_model_size <- which.min(mean_cv_errors)
best_model_size  # Optimal model size according to CV

# 5. Refit Full Subset Model on Training Set (for R², Cp, BIC)
regfit <- regsubsets(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +
                     hour + is_holiday + is_weekend + commute +
                     weather_code + month + t1:commute + t1:hum,
                    data = train, nvmax = max_predictors)
```

```
reg_summary <- summary(regfit)

# 6. Plot Adjusted R², Cp, and BIC
par(mfrow = c(1, 3))
plot(reg_summary$adjr2, type = "b", pch = 20,
    xlab = "Number of Predictors", ylab = "Adjusted R²",
    main = "Adjusted R² by Model Size")
points(which.max(reg_summary$adjr2), max(reg_summary$adjr2), col = "red", pch = 19)

plot(reg_summary$cp, type = "b", pch = 20,
    xlab = "Number of Predictors", ylab = "Cp",
    main = "Cp by Model Size")
points(which.min(reg_summary$cp), min(reg_summary$cp), col = "blue", pch = 19)

plot(reg_summary$bic, type = "b", pch = 20,
    xlab = "Number of Predictors", ylab = "BIC",
    main = "BIC by Model Size")
points(which.min(reg_summary$bic), min(reg_summary$bic), col = "green", pch = 19)
par(mfrow = c(1, 1))

# Best model size by Adjusted R²
which.max(reg_summary$adjr2)
```

```
# Best model size by Cp
which.min(reg_summary$cp)

# Best model size by BIC
which.min(reg_summary$bic)

# 7. View Final Selected Predictors
best_subset_final <- regfit
coef(best_subset_final, id = best_model_size)

# 8. Refit Final Model Based on Selected Predictors
model_subset <- lm(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +
                   hour + is_holiday + is_weekend + commute +
                   weather_code + month +
                   t1:commute + t1:hum,
                  data = train)
summary(model_subset)

# 9. Predict on Test Set and Compute Test MSE
subset_preds <- predict(model_subset, newdata = test)
subset_mse <- mean((test$cnt - subset_preds)^2)
```

```
subset_mse


# 10. Residual Diagnostics

par(mfrow = c(2, 2))

plot(model_subset)



# LASSO Regression with Cross-Validation

# Load glmnet

library(glmnet)


# Build Model Matrix (for LASSO, must use model.matrix to handle factor dummies)

x_train <- model.matrix(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +

                        hour + is_holiday + is_weekend + commute +

                        weather_code + month +

                        t1:commute + t1:hum,

                       data = train)[, -1]  # Remove intercept column


y_train <- train$cnt


x_test <- model.matrix(cnt ~ t1 + I(t1^2) + log(hum + 1) + sqrt(wind_speed) +

                        hour + is_holiday + is_weekend + commute +
```

```
                  weather_code + month +
                  t1:commute + t1:hum,
                 data = test)[, -1]

y_test <- test$cnt

# Run 10-Fold Cross-Validated LASSO
set.seed(123)
cv.lasso <- cv.glmnet(x_train, y_train, alpha = 1, nfolds = 10)
plot(cv.lasso)

# Best lambda
best_lambda <- cv.lasso$lambda.min
best_lambda

# Fit Final LASSO Model Using Best Lambda
lasso_model <- glmnet(x_train, y_train, alpha = 1, lambda = best_lambda)

# Get coefficients of selected predictors
coef(lasso_model)

# Predict on Test Set & Calculate Test Error
```

```
lasso_preds <- predict(lasso_model, s = best_lambda, newx = x_test)

lasso_mse <- mean((y_test - lasso_preds)^2)

lasso_mse


# Residual Diagnostics Plot

lasso_resid <- y_test - lasso_preds


plot(lasso_preds, lasso_resid,

    main = "LASSO Residuals vs Fitted Values",

    xlab = "Fitted Values", ylab = "Residuals", pch = 20)

abline(h = 0, col = "red", lty = 2)


# Classification Problem

# Split Data

set.seed(456)  # Different from regression seed to keep splits independent

n <- nrow(Bike)

train_idx_class <- sample(1:n, size = floor(0.7 * n))

train_class <- Bike[train_idx_class, ]

test_class <- Bike[-train_idx_class, ]


# Logistic Regression Model

# Step 1: Fit The Model on the Training Data
```

```
log_model <- glm(high_demand ~ t1 + hum + hour + is_holiday + is_weekend +
                 commute + t1:commute + t1:hum,
                data = train_class,
                family = "binomial")
summary(log_model)



# Step 2: Predict on Test Set (Get Probabilities & Class Labels)
# Predict probabilities
log_probs <- predict(log_model, newdata = test_class, type = "response")

# Convert to 0/1 using 0.5 threshold
log_preds <- ifelse(log_probs > 0.5, 1, 0)

# Step 3: Confusion Matrix & Metrics (Accuracy, Sensitivity, Specificity)
# Convert to factors with defined levels
actual <- factor(test_class$high_demand, levels = c(0, 1))
predicted <- factor(log_preds, levels = c(0, 1))
conf_matrix <- table(Predicted = predicted, Actual = actual)
conf_matrix

# Sensitivity = TP / (TP + FN)
```

```
sensitivity_log <- conf_matrix["1", "1"] / sum(conf_matrix[, "1"])

sensitivity_log

# Specificity = TN / (TN + FP)

specificity_log <- conf_matrix["0", "0"] / sum(conf_matrix[, "0"])

specificity_log

# Accuracy (optional)

accuracy_log <- mean(predicted == actual)

accuracy_log

# Test error (misclassification rate)

test_error_log <- mean(predicted != actual)

test_error_log


# LDA Model for Classification

# 1 Load Mass and Fit the Model

library(MASS)

lda_model <- lda(high_demand ~ t1 + hum + hour + is_holiday + is_weekend +

                 commute + t1:commute + t1:hum,

               data = train_class)

lda_model



# 2 Predict on Test Set
```

```
lda_pred <- predict(lda_model, newdata = test_class)

lda_class <- lda_pred$class  # Predicted class labels


# 3 Confusion Matrix and Metrics

# Convert to factors with same levels

actual_lda <- factor(test_class$high_demand, levels = c(0, 1))

predicted_lda <- factor(lda_class, levels = c(0, 1))


conf_matrix_lda <- table(Predicted = predicted_lda, Actual = actual_lda)

conf_matrix_lda


# 4: Calculate Accuracy, Test Error, Sensitivity, Specificity

# Confusion matrix elements

TP_lda <- conf_matrix_lda["1", "1"]

TN_lda <- conf_matrix_lda["0", "0"]

FP_lda <- conf_matrix_lda["1", "0"]

FN_lda <- conf_matrix_lda["0", "1"]


# Metrics

accuracy_lda <- (TP_lda + TN_lda) / sum(conf_matrix_lda)

accuracy_lda

test_error_lda <- 1 - accuracy_lda
```

```
test_error_lda

sensitivity_lda <- TP_lda / (TP_lda + FN_lda)

sensitivity_lda

specificity_lda <- TN_lda / (TN_lda + FP_lda)

specificity_lda


# KNN for Classification

# 1 Ensure Numeric Format for KNN

# Convert t1 and hum to numeric (if not already)

train_class$t1 <- as.numeric(as.character(train_class$t1))

train_class$hum <- as.numeric(as.character(train_class$hum))

test_class$t1 <- as.numeric(as.character(test_class$t1))

test_class$hum <- as.numeric(as.character(test_class$hum))


# Convert commute, is_holiday, is_weekend to numeric (0/1)

train_class$commute <- as.numeric(as.character(train_class$commute))

train_class$is_holiday <- as.numeric(as.character(train_class$is_holiday))

train_class$is_weekend <- as.numeric(as.character(train_class$is_weekend))

test_class$commute <- as.numeric(as.character(test_class$commute))

test_class$is_holiday <- as.numeric(as.character(test_class$is_holiday))

test_class$is_weekend <- as.numeric(as.character(test_class$is_weekend))
```

```
# 2 Create Interaction Terms

train_class$`t1:commute` <- train_class$t1 * train_class$commute

train_class$`t1:hum` <- train_class$t1 * train_class$hum


test_class$`t1:commute` <- test_class$t1 * test_class$commute

test_class$`t1:hum` <- test_class$t1 * test_class$hum


# 3 Prepare Scaled Predictor Matrices

library(caret)


knn_vars <- c("t1", "hum", "hour", "is_holiday", "is_weekend", "commute", "t1:commute",
"t1:hum")


# Standardize predictors

preProc <- preProcess(train_class[, knn_vars], method = c("center", "scale"))

train_knn <- predict(preProc, train_class[, knn_vars])

test_knn <- predict(preProc, test_class[, knn_vars])


# Define response variable

train_labels <- train_class$high_demand

test_labels <- test_class$high_demand


# 4 Cross-Validation to Choose Optimal K
```

```
library(class)

set.seed(123)

# Try a range of odd K values
k_values <- seq(1, 25, by = 2)

cv_errors <- sapply(k_values, function(k) {
  preds <- knn(train = train_knn, test = test_knn, cl = train_labels, k = k)
  mean(preds != test_labels)
})

# Plot test errors by K
plot(k_values, cv_errors, type = "b", pch = 19,
     xlab = "K (Number of Neighbors)", ylab = "Test Error",
     main = "KNN Cross-Validation for Optimal K")

# 5 Fit Final KNN Model Using Optimal K
best_k <- k_values[which.min(cv_errors)]
best_k

# Final KNN model
```

```
knn_preds <- knn(train = train_knn, test = test_knn, cl = train_labels, k = best_k)


# 6 Confusion Matrix and Metrics

# Convert to factors

actual_knn <- factor(test_labels, levels = c(0, 1))

predicted_knn <- factor(knn_preds, levels = c(0, 1))


conf_matrix_knn <- table(Predicted = predicted_knn, Actual = actual_knn)

conf_matrix_knn


# Extract values

TP_knn <- conf_matrix_knn["1", "1"]

TN_knn <- conf_matrix_knn["0", "0"]

FP_knn <- conf_matrix_knn["1", "0"]

FN_knn <- conf_matrix_knn["0", "1"]


# Metrics

accuracy_knn <- (TP_knn + TN_knn) / sum(conf_matrix_knn)

test_error_knn <- 1 - accuracy_knn

sensitivity_knn <- TP_knn / (TP_knn + FN_knn)

specificity_knn <- TN_knn / (TN_knn + FP_knn)
```

```
# Display results
list(
  best_k = best_k,
  test_error = test_error_knn,
  sensitivity = sensitivity_knn,
  specificity = specificity_knn,
  accuracy = accuracy_knn)
```